BankChurners Data Analysis

PGP DSBA - Guided Project Report

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1. Description:- Machine Learning 2: BankChurners Data Analysis:-

Business Context

The Thera bank recently saw a steep decline in the number of users of their credit card, credit cards are a good source of income for banks because of different kinds of fees charged by the banks like annual fees, balance transfer fees, and cash advance fees, late payment fees, foreign transaction fees, and others. Some fees are charged to every user irrespective of usage, while others are charged under specified circumstances.

Customers' leaving credit cards services would lead bank to loss, so the bank wants to analyze the data of customers and identify the customers who will leave their credit card services and reason for same – so that bank could improve upon those areas.

The bank need to come up with a classification model that will help the bank improve its services so that customers do not renounce their credit cards.

2. Data Description

The data contains the different attributes of leads and their interaction details with BankChurners. The detailed data dictionary is given below.

2.1 Data Dictionary

- CLIENTNUM: Client number. Unique identifier for the customer holding the account
- Attrition_Flag: Internal event (customer activity) variable - if the account is closed then "Attrited Customer" else "Existing Customer"
- Customer_Age: Age in Years
- Gender: Gender of the account holder
- Dependent_count: Number of dependents
- Education_Level: Educational Qualification of the account holder - Graduate, High School, Unknown, Uneducated, College(refers to college student), Post-Graduate, Doctorate
- Marital_Status: Marital Status of the account holder
- Income_Category: Annual Income Category of the account holder
- Card_Category: Type of Card
- Months_on_book: Period of relationship with the bank (in months)
- Total_Relationship_Count: Total no. of products held by the customer
- Months_Inactive_12_mon: No. of months inactive in the last 12 months
- Contacts_Count_12_mon: No. of Contacts in the last 12 months
- Credit_Limit: Credit Limit on the Credit Card
- Total_Revolving_Bal: Total Revolving Balance on the Credit Card
- Avg_Open_To_Buy: Open to Buy Credit Line (Average of last 12 months)
- Total_Amt_Chng_Q4_Q1: Change in Transaction Amount (Q4 over Q1)

- Total_Trans_Amt: Total Transaction Amount (Last 12 months)
- Total_Trans_Ct: Total Transaction Count (Last 12 months)
- Total_Ct_Chng_Q4_Q1: Change in Transaction Count (Q4 over Q1)
- Avg_Utilization_Ratio: Average Card Utilization Ratio
- What Is a Revolving Balance?

  If we don't pay the balance of the revolving credit account in full every month, the unpaid portion carries over to the next month. That's called a revolving balance

- What is the Average Open to buy?

  'Open to Buy' means the amount left on your credit card to use. Now, this column represents the average of this value for the last 12 months.

- What is the Average utilization Ratio?

  The Avg_Utilization_Ratio represents how much of the available credit the customer spent. This is useful for calculating credit scores.

- Relation b/w Avg_Open_To_Buy, Credit_Limit and Avg_Utilization_Ratio:

  (Avg_Open_To_Buy / Credit_Limit) + Avg_Utilization_Ratio = 1

3. Data Overview

3.1 Checking the shape of the dataset:-

The dataset contains information (21 attributes) about 10127 CLIENTNUM rows.

3.2 Displaying the first five rows of the dataset:-

- The data frame has 21 columns as mentioned in the Data Dictionary.
- Data in each row corresponds to the leads and their interaction details with BankChurners.

Row No.	CLIENTNUM	Attrition_Flag	Customer_Age	Gender	Dependent_count	Education_Level	Marital_Status	Income_Category	Card_Category	Months_on_book
0	768805383	Existing Customer	45	M	3	High School	Married	$60K - $80K	Blue	39
1	818770008	Existing Customer	49	F	5	Graduate	Single	Less than $40K	Blue	44
2	713982108	Existing Customer	51	M	3	Graduate	Married	$80K - $120K	Blue	36
3	769911858	Existing Customer	40	F	4	High School	NaN	Less than $40K	Blue	34
4	709106358	Existing Customer	40	M	3	Uneducated	Married	$60K - $80K	Blue	21

Total_Relationship_Count	Months_Inactive_12_mon	Contacts_Count_12_mon	Credit_Limit	Total_Revolving_Bal	Avg_Open_To_Buy	Total_Amt_Chng_Q4_Q1	Total_Trans_Amt	Total_Trans_Ct	Total_Ct_Chng_Q4_Q1	Avg_Utilization_Ratio
5	1	3	12691	777	11914	1.335	1144	42	1.625	0.06
6	1	2	8256	864	7392	1.541	1291	33	3.714	0.11
4	1	0	3418	0	3418	2.594	1887	20	2.333	0
3	4	1	3313	2517	796	1.405	1171	20	2.333	0.76
5	1	0	4716	0	4716	2.175	816	28	2.5	0

Table 1 list of top 5 row

3.3 View the last 5 rows of the dataset:-

- It is to view the last five rows of the dataset.

Row No.	CLIENTNUM	Attrition_Flag	Customer_Age	Gender	Dependent_count	Education_Level	Marital_Status	Income_Category	Card_Category	Months_on_book
10122	772366833	Existing Customer	50	M	2	Graduate	Single	$40K - $60K	Blue	40
10123	710638233	Attrited Customer	41	M	2	NaN	Divorced	$40K - $60K	Blue	25
10124	716506083	Attrited Customer	44	F	1	High School	Married	Less than $40K	Blue	36
10125	717406983	Attrited Customer	30	M	2	Graduate	NaN	$40K - $60K	Blue	36
10126	714337233	Attrited Customer	43	F	2	Graduate	Married	Less than $40K	Silver	25

Total_Relationship_Count	Months_Inactive_12_mon	Contacts_Count_12_mon	Credit_Limit	Total_Revolving_Bal	Avg_Open_To_Buy	Total_Amt_Chng_Q4_Q1	Total_Trans_Amt	Total_Trans_Ct	Total_Ct_Chng_Q4_Q1	Avg_Utilization_Ratio
3	2	3	4003	1851	2152	0.703	15476	117	0.857	0.462
4	2	3	4277	2186	2091	0.804	8764	69	0.683	0.511
5	3	4	5409	0	5409	0.819	10291	60	0.818	0
4	3	3	5281	0	5281	0.535	8395	62	0.722	0
6	2	4	10388	1961	8427	0.703	10294	61	0.649	0.189

Table 2 Last 5 rows of data set

3.4 Check the data types of the columns for the dataset.

- The column range index has 10127 entries (i.e No.of rows = 10127) which starts the number series from 0 to 10126.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10127 entries, 0 to 10126
Data columns (total 21 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   CLIENTNUM                 10127 non-null  int64
 1   Attrition_Flag            10127 non-null  object
 2   Customer_Age              10127 non-null  int64
 3   Gender                    10127 non-null  object
 4   Dependent_count           10127 non-null  int64
 5   Education_Level           8608 non-null   object
 6   Marital_Status            9378 non-null   object
 7   Income_Category           10127 non-null  object
 8   Card_Category             10127 non-null  object
 9   Months_on_book            10127 non-null  int64
 10  Total_Relationship_Count  10127 non-null  int64
 11  Months_Inactive_12_mon    10127 non-null  int64
 12  Contacts_Count_12_mon     10127 non-null  int64
 13  Credit_Limit              10127 non-null  float64
 14  Total_Revolving_Bal       10127 non-null  int64
 15  Avg_Open_To_Buy           10127 non-null  float64
 16  Total_Amt_Chng_Q4_Q1      10127 non-null  float64
 17  Total_Trans_Amt           10127 non-null  int64
 18  Total_Trans_Ct            10127 non-null  int64
 19  Total_Ct_Chng_Q4_Q1       10127 non-null  float64
 20  Avg_Utilization_Ratio     10127 non-null  float64
dtypes: float64(5), int64(10), object(6)
memory usage: 1.6+ MB
```

Table 3 Data type information

3.5 Checking the duplicate values

- There are 0 (i.e. zero) duplicate values.

	0
0	False
1	False
2	False
3	False
4	False
...	...
10122	False
10123	False
10124	False
10125	False
10126	False

10127 rows × 1 columns

dtype: bool

Table 4 Duplicate values

3.6 Checking the missing values

The data set has missing values for Education_Level and Marital_Status attributes remaining have no missing values.

Education_Level has 1519 missing values and Marital_Status has 749 missing values.

	0
CLIENTNUM	0
Attrition_Flag	0
Customer_Age	0
Gender	0
Dependent_count	0
Education_Level	1519
Marital_Status	749
Income_Category	0
Card_Category	0
Months_on_book	0
Total_Relationship_Count	0
Months_Inactive_12_mon	0
Contacts_Count_12_mon	0
Credit_Limit	0
Total_Revolving_Bal	0
Avg_Open_To_Buy	0
Total_Amt_Chng_Q4_Q1	0
Total_Trans_Amt	0
Total_Trans_Ct	0
Total_Ct_Chng_Q4_Q1	0
Avg_Utilization_Ratio	0

dtype: int64

Table 5 Missing values

3.7 Checking the statistical summary

- This table reflects the statistical summary of the data contains - Count, Mean, Std (Standard deviation), Min (Minimum), Max (Maximum), and Percentages (25%, 50%, 75%).

	count	mean	std	min	25%	50%	75%	max
CLIENTNUM	10127.000	739177606.334	36903783.450	708082083.000	713036770.500	717926358.000	773143533.000	828343083.000
Customer_Age	10127.000	46.326	8.017	26.000	41.000	46.000	52.000	73.000
Dependent_count	10127.000	2.346	1.299	0.000	1.000	2.000	3.000	5.000
Months_on_book	10127.000	35.928	7.986	13.000	31.000	36.000	40.000	56.000
Total_Relationship_Count	10127.000	3.813	1.554	1.000	3.000	4.000	5.000	6.000
Months_Inactive_12_mon	10127.000	2.341	1.011	0.000	2.000	2.000	3.000	6.000
Contacts_Count_12_mon	10127.000	2.455	1.106	0.000	2.000	2.000	3.000	6.000
Credit_Limit	10127.000	8631.954	9088.777	1438.300	2555.000	4549.000	11067.500	34516.000
Total_Revolving_Bal	10127.000	1162.814	814.987	0.000	359.000	1276.000	1784.000	2517.000
Avg_Open_To_Buy	10127.000	7469.140	9090.685	3.000	1324.500	3474.000	9859.000	34516.000
Total_Amt_Chng_Q4_Q1	10127.000	0.760	0.219	0.000	0.631	0.736	0.859	3.397
Total_Trans_Amt	10127.000	4404.086	3397.129	510.000	2155.500	3899.000	4741.000	18484.000
Total_Trans_Ct	10127.000	64.859	23.473	10.000	45.000	67.000	81.000	139.000
Total_Ct_Chng_Q4_Q1	10127.000	0.712	0.238	0.000	0.582	0.702	0.818	3.714
Avg_Utilization_Ratio	10127.000	0.275	0.276	0.000	0.023	0.176	0.503	0.999

Table 6 Statistical Summary of the data set

3.8 Checking the statistical summary for the Object category:-

- This table reflects the statistical summary of the data contains object category - Count, unique and top freq.

	count	unique	top	freq
Attrition_Flag	10127	2	Existing Customer	8500
Gender	10127	2	F	5358
Education_Level	8608	6	Graduate	3128
Marital_Status	9378	3	Married	4687
Income_Category	10127	6	Less than $40K	3561
Card_Category	10127	4	Blue	9436

Table 7 Object data type statistical summary

3.8 Checking the number of unique values

- The number of unique values of various attritions/Columns.

```
Unique values in Attrition_Flag are :
Attrition_Flag
Existing Customer    8500
Attrited Customer    1627
Name: count, dtype: int64
**************************************************
Unique values in Gender are :
Gender
F    5358
M    4769
Name: count, dtype: int64
**************************************************
Unique values in Education_Level are :
Education_Level
Graduate         3128
High School      2013
Uneducated       1487
College          1013
Post-Graduate     516
Doctorate         451
Name: count, dtype: int64
**************************************************
Unique values in Marital_Status are :
Marital_Status
Married     4687
Single      3943
Divorced     748
Name: count, dtype: int64
**************************************************
Unique values in Income_Category are :
Income_Category
Less than $40K    3561
$40K - $60K       1790
$80K - $120K      1535
$60K - $80K       1402
abc               1112
$120K +            727
Name: count, dtype: int64
**************************************************
Unique values in Card_Category are :
Card_Category
Blue        9436
Silver       555
Gold         116
Platinum      20
Name: count, dtype: int64
**************************************************
```

Table 8 Unique values

3.9 CLIENTNUM after drop from the data set

These are available columns after drop CLIENTNUM from the existing dataset.

```
Index(['Attrition_Flag', 'Customer_Age', 'Gender', 'Dependent_count',
       'Education_Level', 'Marital_Status', 'Income_Category', 'Card_Category',
       'Months_on_book', 'Total_Relationship_Count', 'Months_Inactive_12_mon',
       'Contacts_Count_12_mon', 'Credit_Limit', 'Total_Revolving_Bal',
       'Avg_Open_To_Buy', 'Total_Amt_Chng_Q4_Q1', 'Total_Trans_Amt',
       'Total_Trans_Ct', 'Total_Ct_Chng_Q4_Q1', 'Avg_Utilization_Ratio'],
      dtype='object')
```

Table 9 Columns after dropped the CLIENTNUM

Exploratory Data Analysis

- EDA has a key role in data analysis for the data science. It includes different types of analysis like Univariate analysis & Bivariate analysis which are explained in the below points.

4. Univariate Analysis:-

It means one variable analysis with count. It refers to the observations of different variables.

4.1 Customer_Age:-

It is to create a histogram_boxplot for 'Observations on age'.

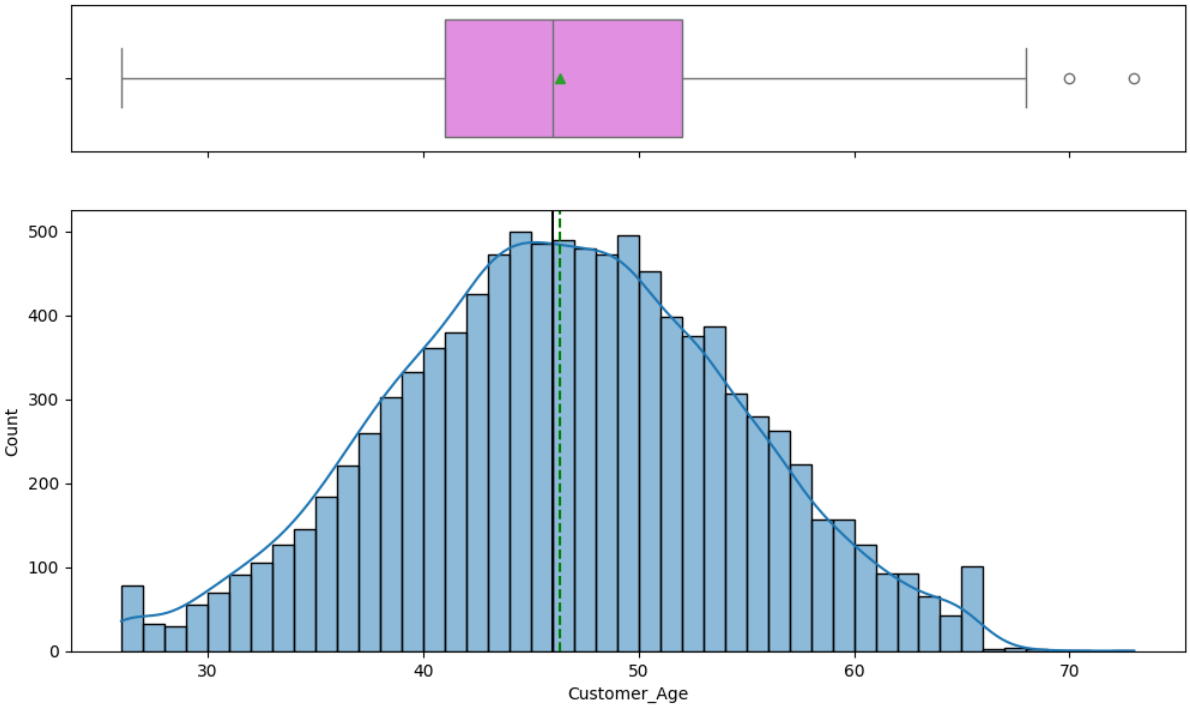

Figure 1 Customer Age

- The customer age observations are approximately 46.32 (Approximately) on average.
- It has two outliers on the right-hand side and it has the right side skewed.
- The median is 46. It is close to normal distribution.

4.2 Months_on_book

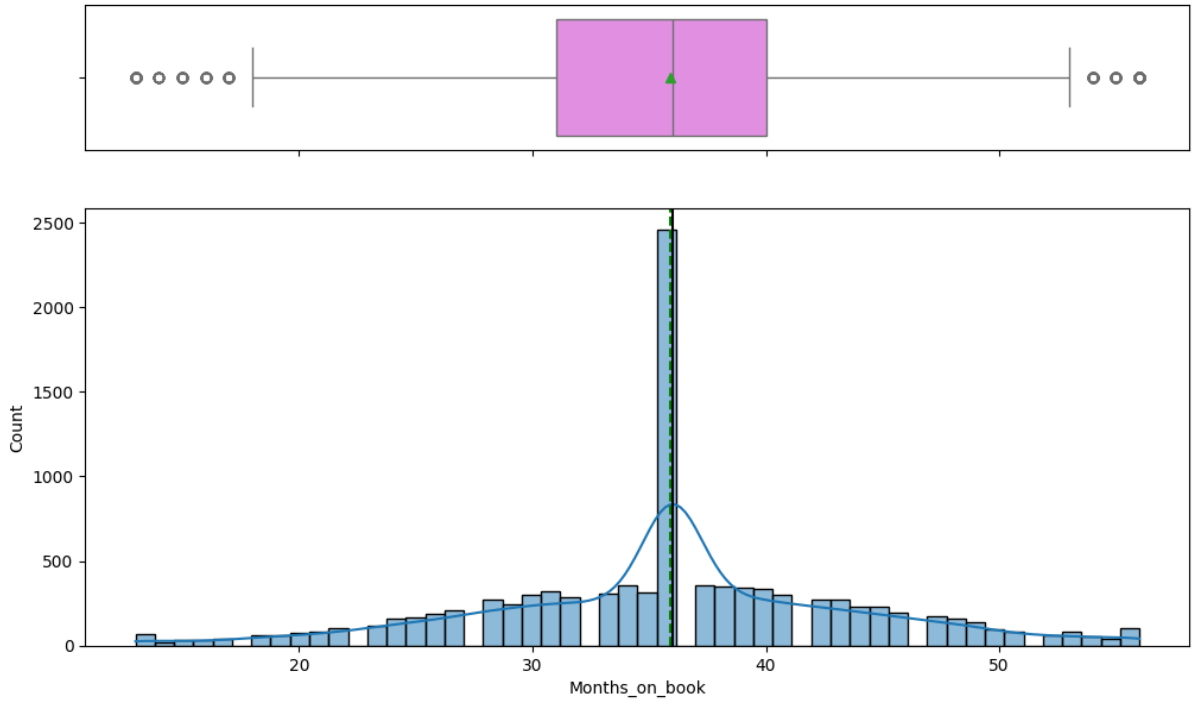

Figure 2 Months on book

- The Months_on_book observations are approximately 35.92 (Approximately) on average.
- It has three outliers on the right-hand side and 5 outliers on the left-hand side.
- The median is 36.

4.3 Credit_Limit

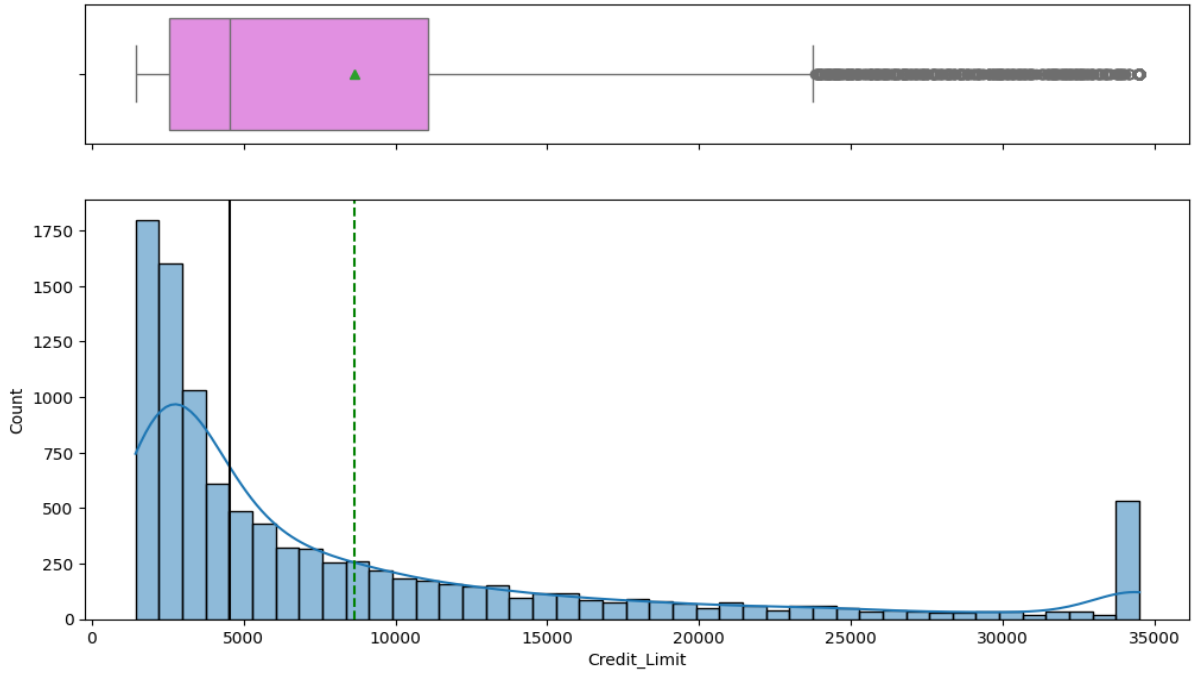

Figure 3 Credit Limit

- The Credit_Limit observations are approximately 8631.95 (Approximately) on average.

- It has many outliers on the right-hand side and it is right skewed.
- The median is 4549.

Total_Revolving_Bal

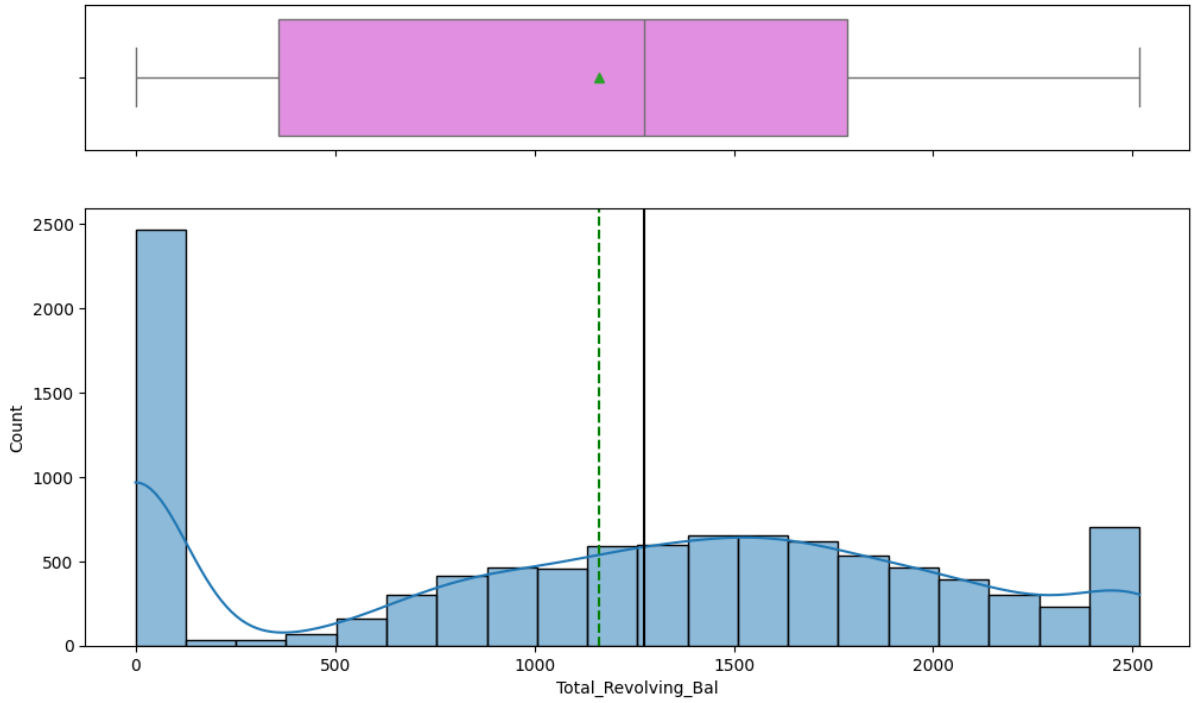

Figure 4 Total Revolving Bal

- The Total_Revolving_Bal observations are approximately 1162.81 (Approximately) on average.
- It has no outliers.
- The median is 1276.

4.4 Avg_Open_To_Buy

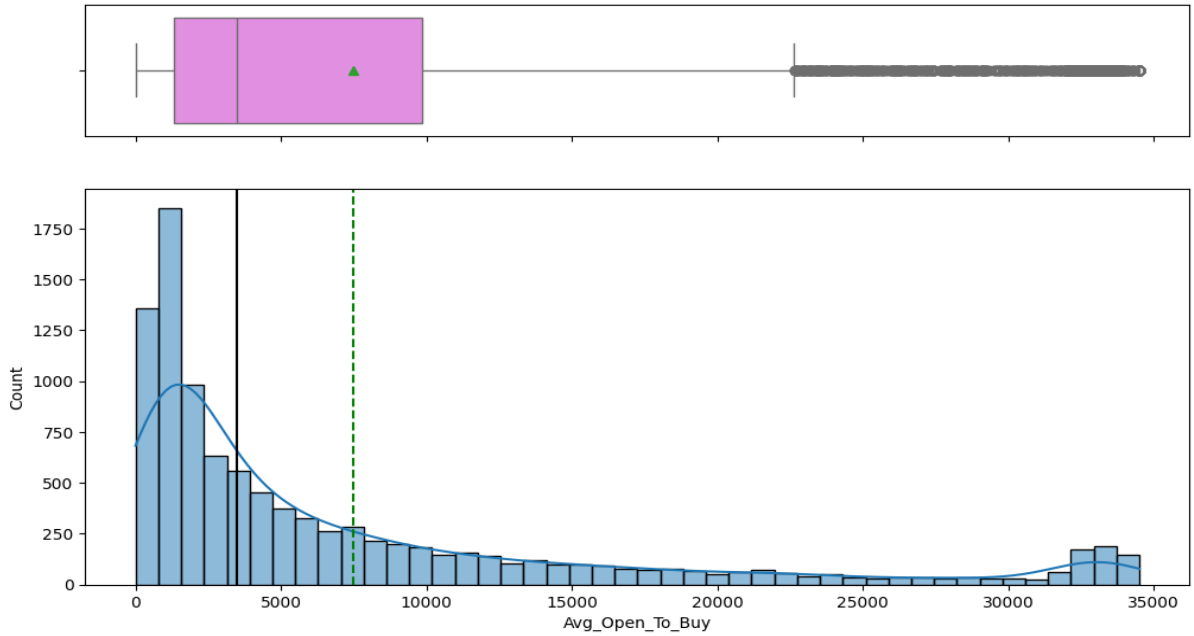

Figure 5 Avg Open to Buy

- The Avg_Open_To_Buy observations are distributed approximately 7469.14 on average.
- It has many outliers on the right-hand side and it is right skewed.
- The median is 3474.

4.5 Total_Trans_Ct

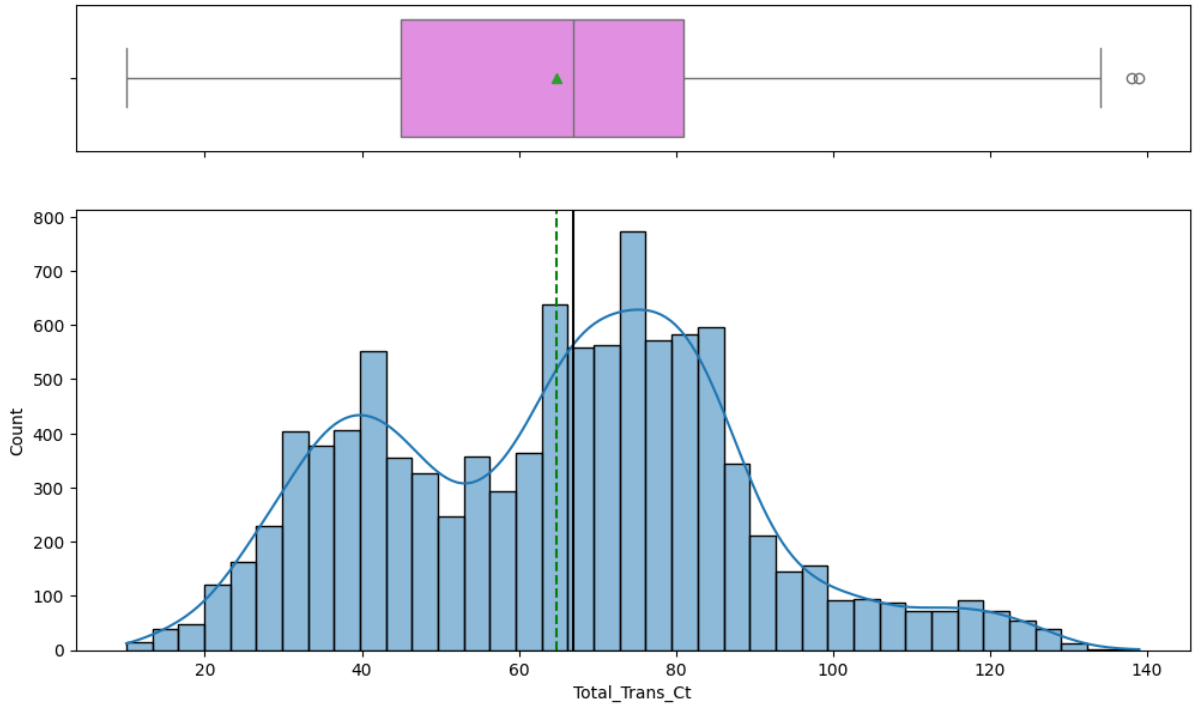

Figure 6 Total Trans Ct

- The Total_Trans_Ct observations are distributed approximately 64.85 on average.
- It has two outliers on the right-hand side and is slightly right-skewed.
- The median is 67.

4.6 Total_Amt_Chng_Q4_Q1

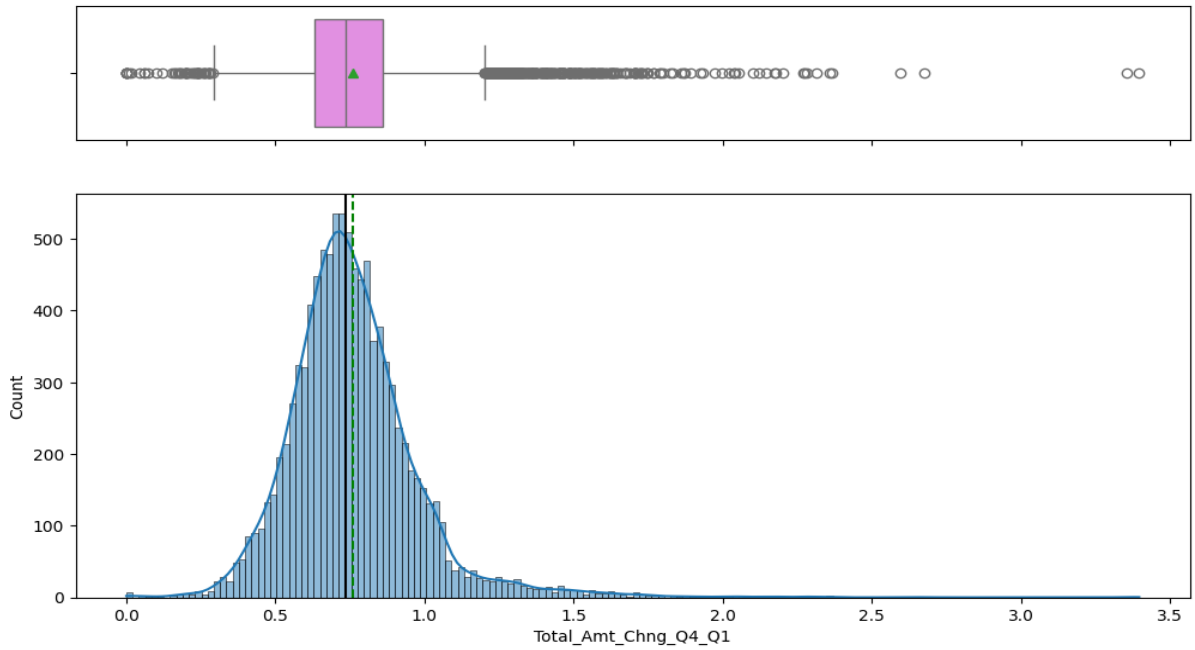

Figure 7 Total Amt Chng Q4 _ Q1

- The Total_Amt_Chng_Q4_Q1 observations are distributed approximately 0.75 on average.
- It has too many outliers on the right-hand side and left-hand side. It is close to normal distribution.
- The median is 0.73 (Approximately).

4.7 Total_Trans_Amt

Let's see the total transaction amount distributed.

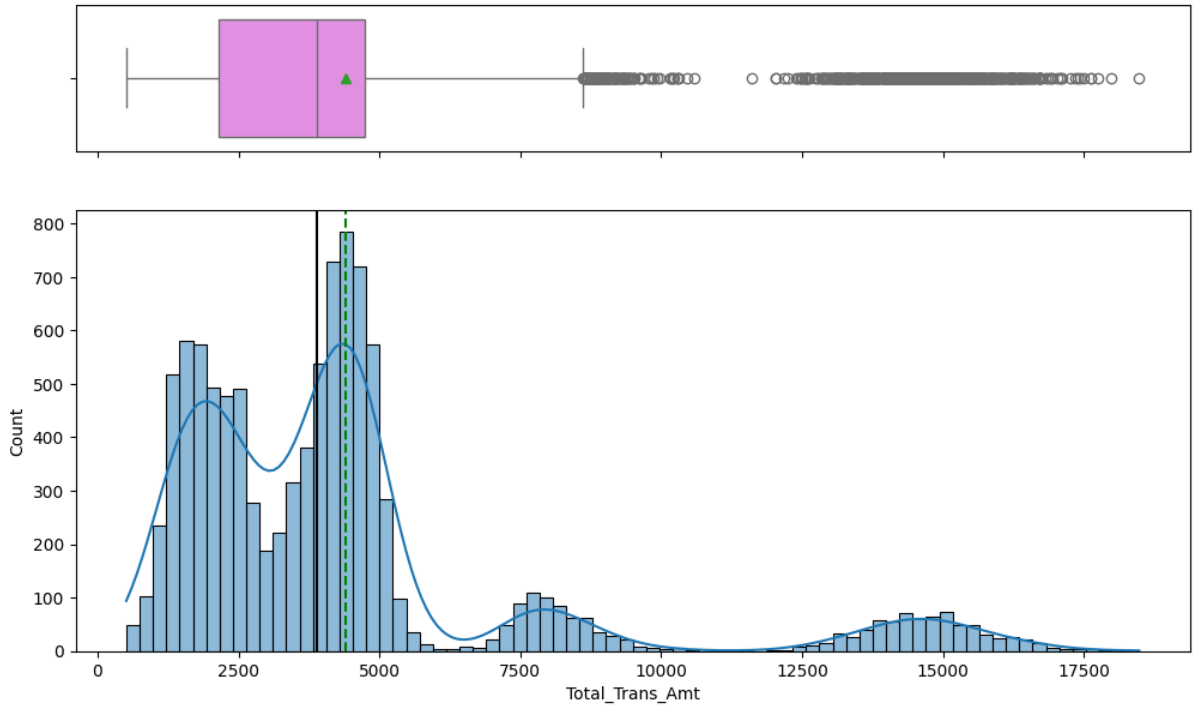

Figure 8 Total Trans Amt

- The Total_Trans_Amt observations are distributed approximately 4404 on average.
- It has too many outliers on the right-hand side and it is right skewed.
- The median is 3899.

4.8 Avg_Utilization_Ratio

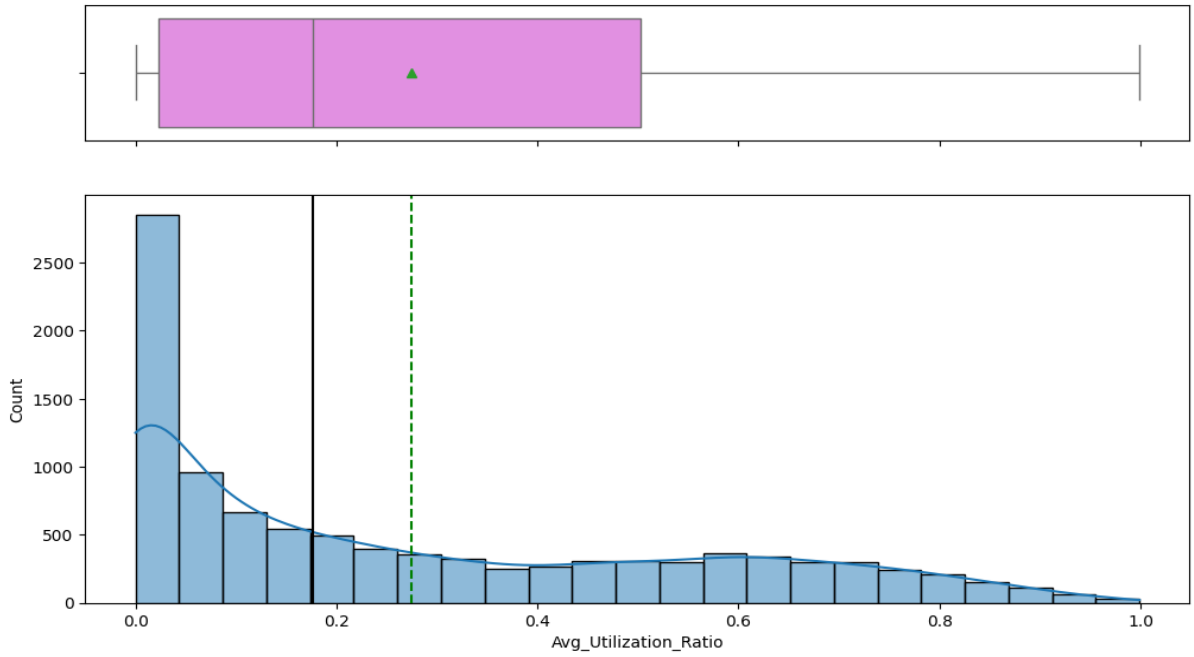

Figure 9 Avg Utilization Ratio

- The Avg_Utilization_Ratio observations are distributed approximately 0.27 on average.
- It has no outliers and it is right skewed.
- The median is 0.176.

4.9 Dependent_count

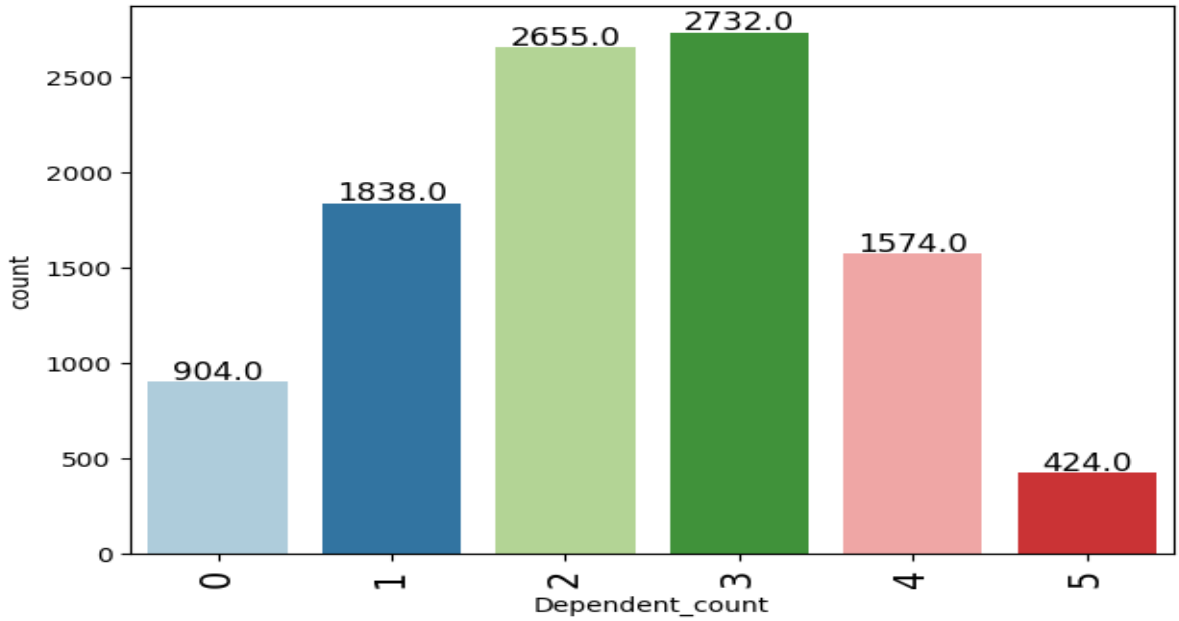

Figure 10 Dependent Count

- The number of 3 dependents is the highest with a dependent_count of 2732 and following the dependents 2 with a dependent_count is 2655.
- Dependents - 5 is the least one with a dependent count of 424.

4.10 Total_Relationship_Count

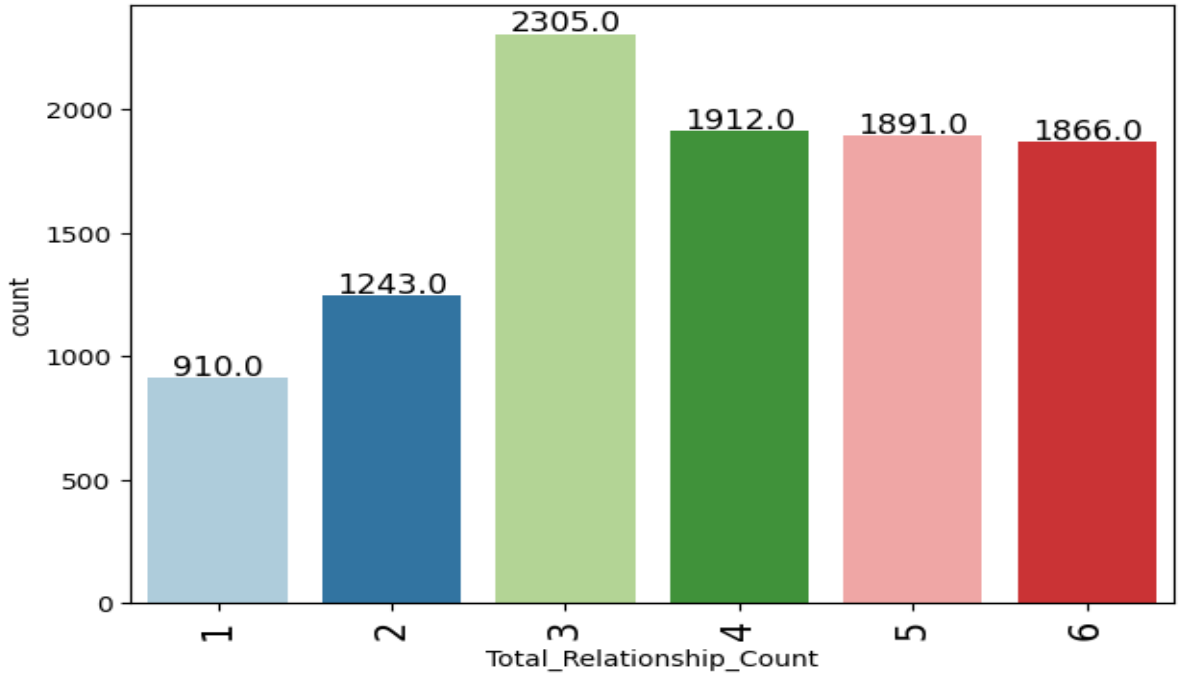

Figure 11 Total Relationship Count

- The total number of products held by the customer is high with a Total_Relationship_Count of 2305 for category - 3.
- The total noumber of products held by the customer is the least for category-1 of Total_Relationship_Count of 910.

4.11 Months_Inactive_12_mon

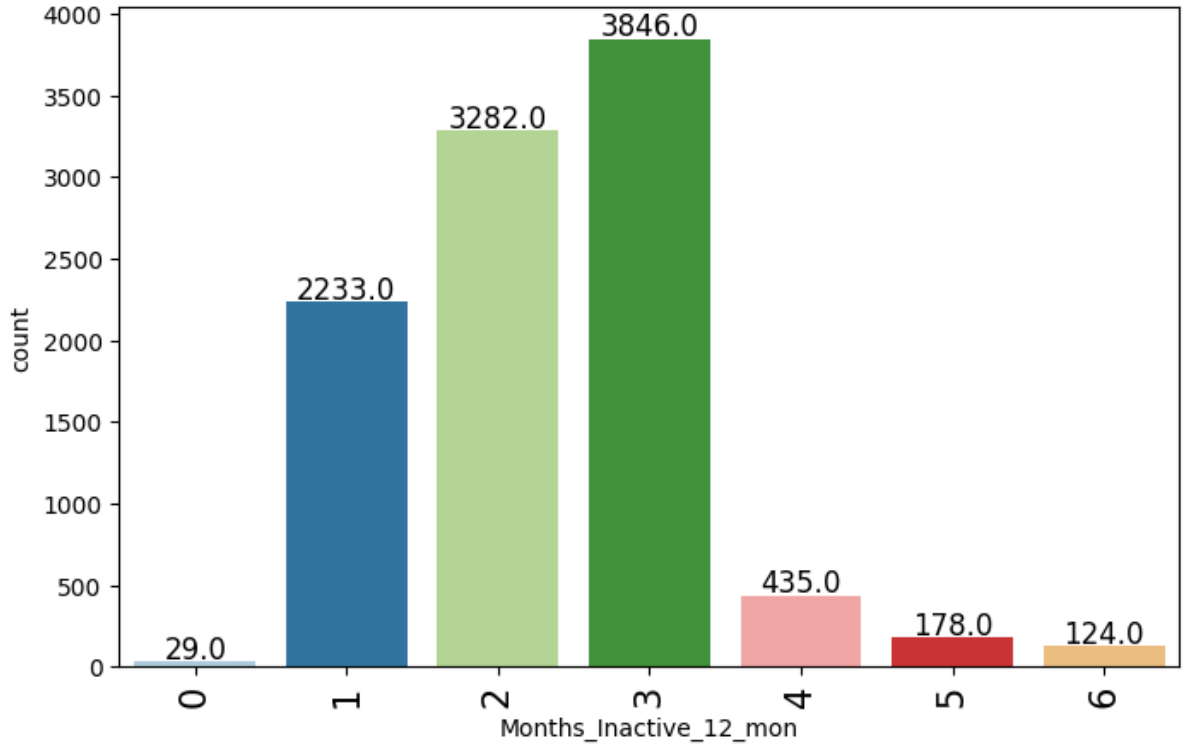

Figure 12 Months Inactive 12 mon

- 3-Months have been inactive in the last 12 months with a customer count of 3846.
- 0-Months are inactive in the last 12-month period with the lowest count of 29.

4.12 Contacts_Count_12_mon

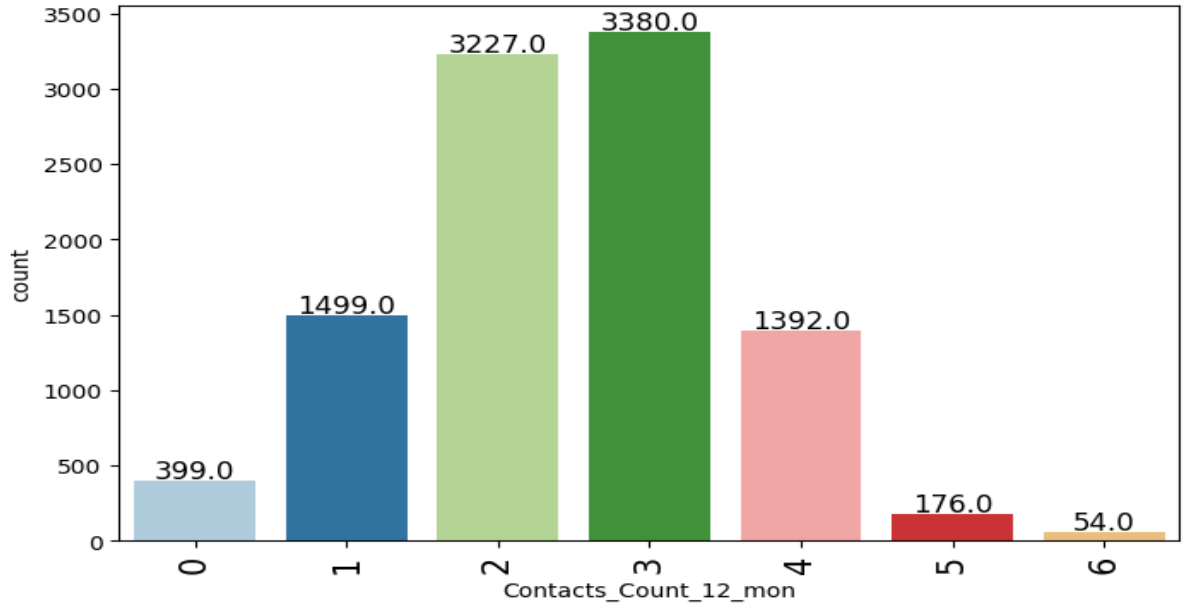

Figure 13 Contacts count 12 mon

- Three contacts in the last 12 months category high with 3380 records.
- Six contacts in the last 12 months have the lowest with 54 records out of all the records of contacts_Count_12_mon.

4.13 Gender

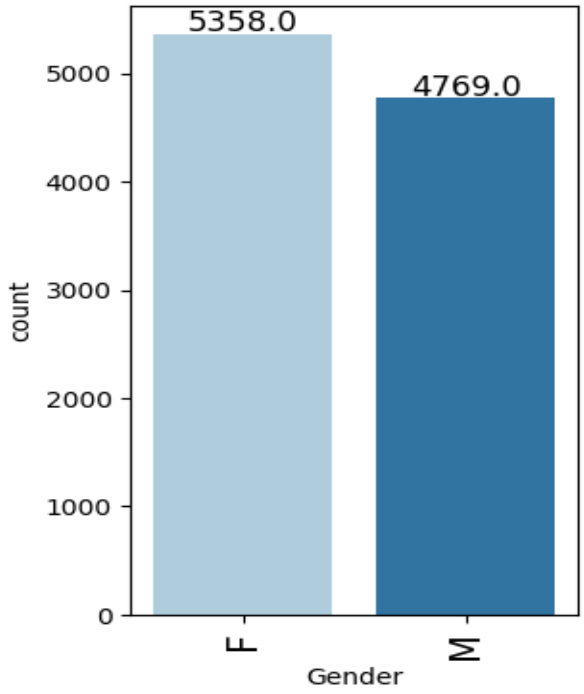

Figure 14 Gender wise count

- Male gender account holders are 4769 which is the least.
- The female gender for account holders is 5358 which is the highest.

4.14 Education_Level

Let's see the distribution of the level of education of customers

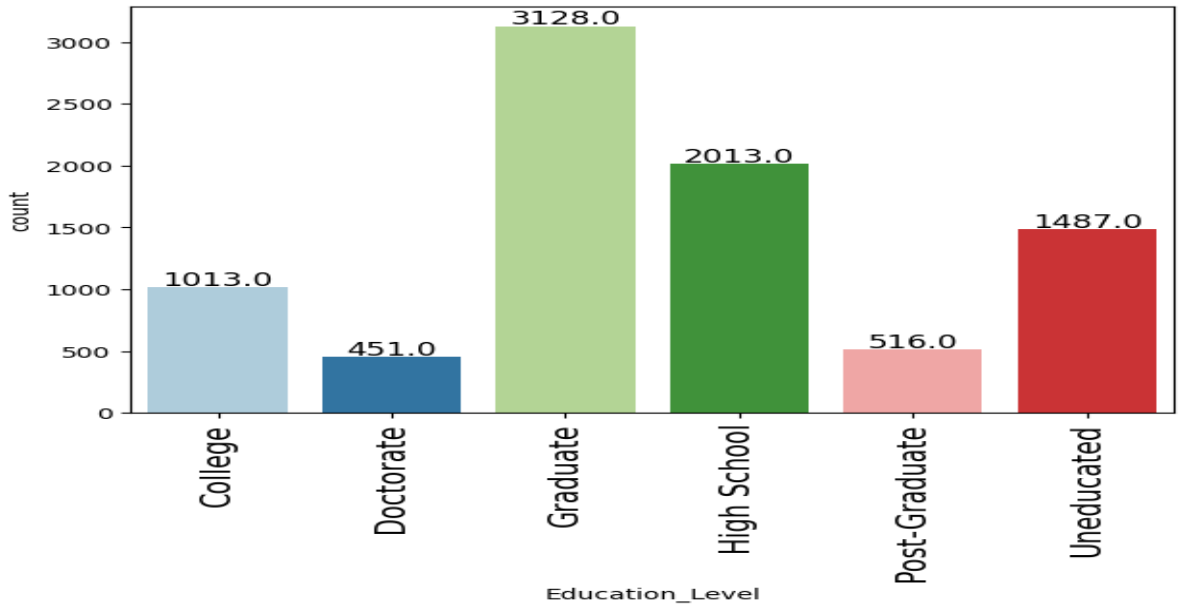

Figure 15 Education Level

- Educational Qualification of the Accountholder – Graduates have the highest records.
- Educational Qualification of the account holder – Doctorates have less number of records.

4.15 Marital_Status

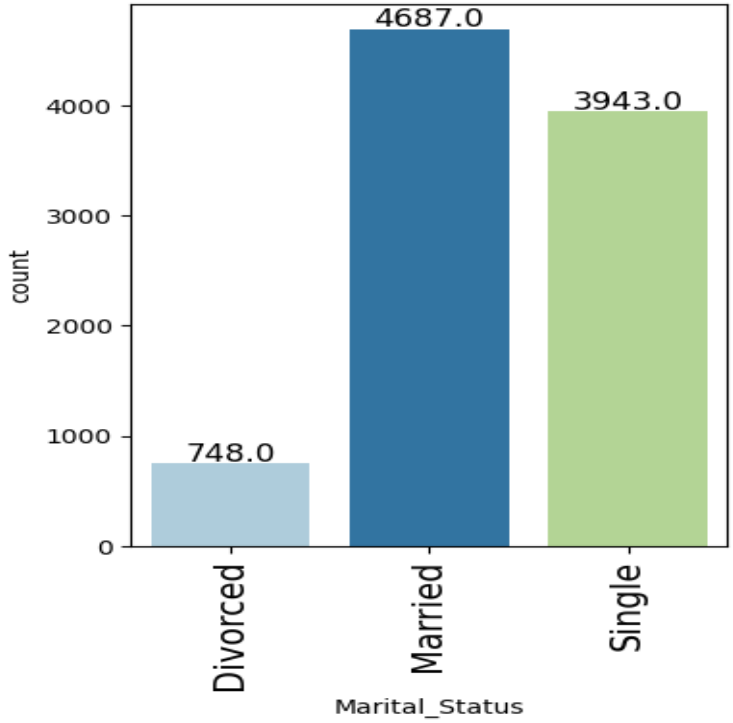

Figure 16 Marital status

- Marital Status of the account holder for the Divorced category has fewer records 748.
- Married - marital status of the account holders is high with 4687 records.

4.16 Income_Category

Let's see the distribution of the level of income of customers

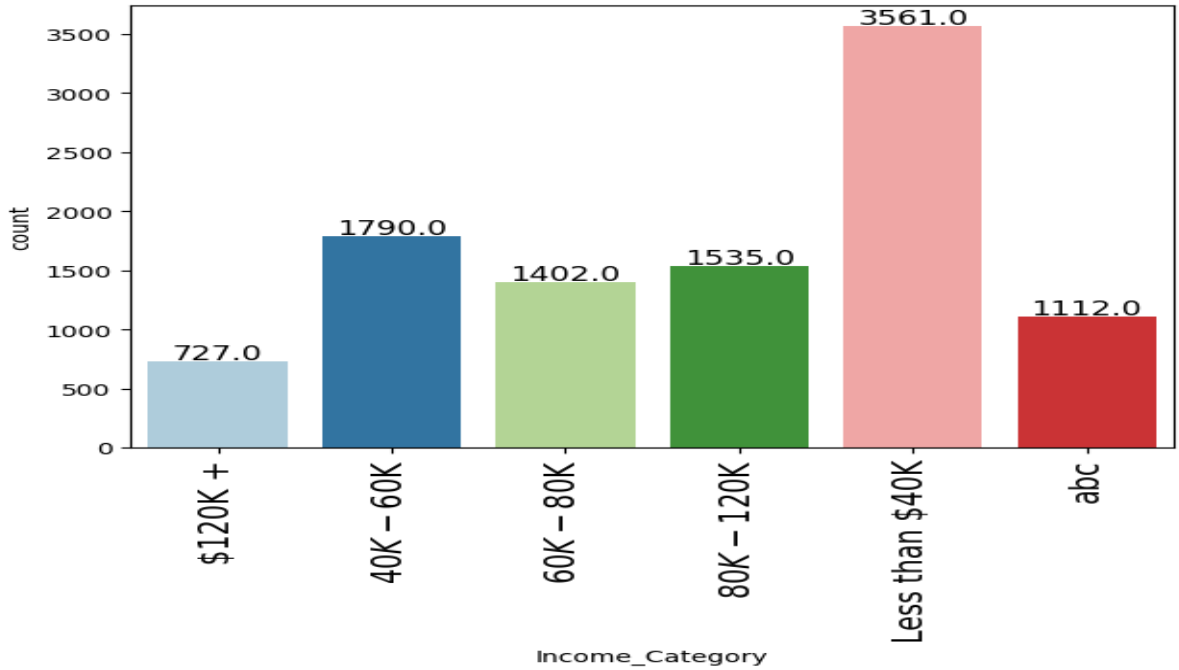

Figure 17 Income Category

- The count of highest annual Income Category of the account holder with less than $40K is 3561.
- The count of lowest annual Income Category of the account holder with $120K is 727.

4.17 Card_Category

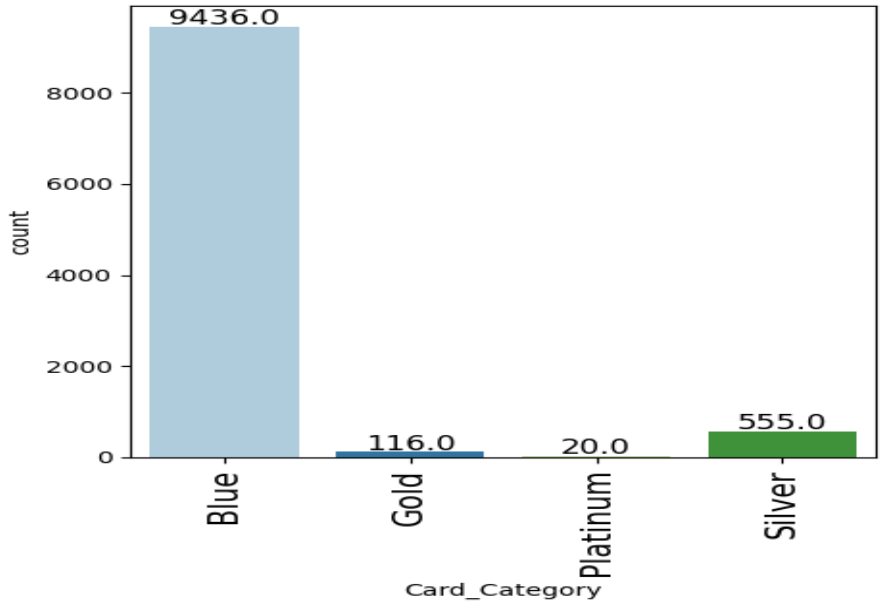

Figure 18 Card Category

- The Blue card category has the highest count 9436.
- The Platinum card category has the lowest count of 20.

4.18 Attrition_Flag

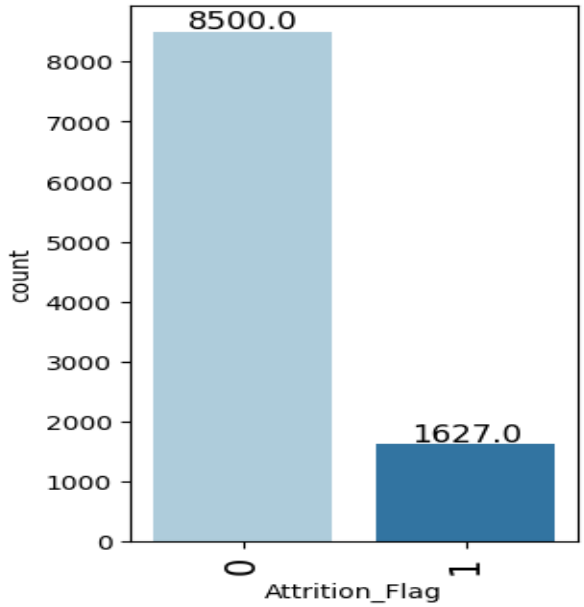

Figure 19 Attrition Flag

- The attrition_Flag -1 has the lowest count of 1627.
- The attrition_Flag -0 has the highest count of 8500.

4.19 Creating histograms:-

This is the histogram for all the categorical data on one screen.

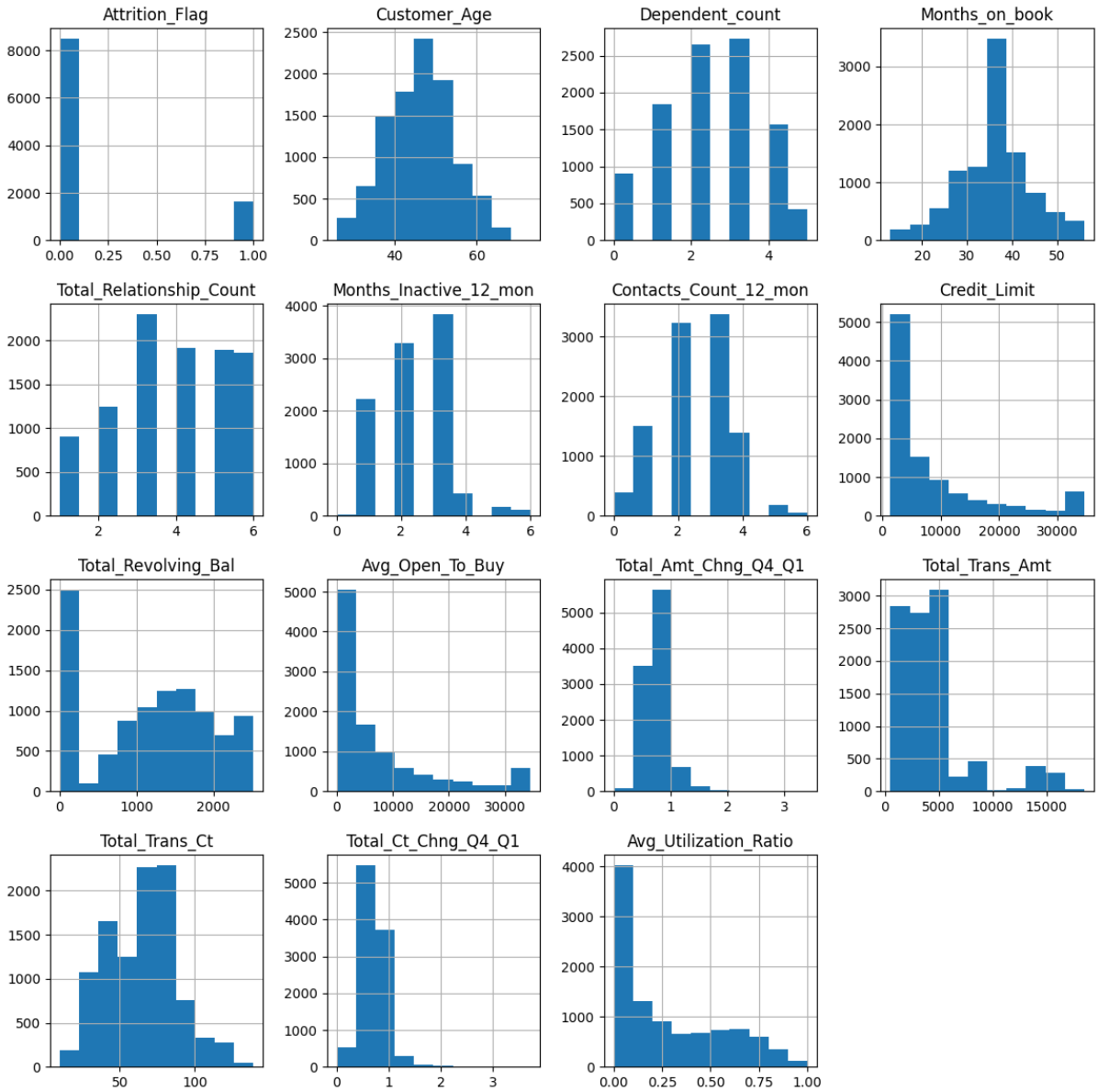

Figure 20 Histograms for categorical data

5. Bivariate Distributions

Let's see the attributes that have a strong correlation with each other.

5.1 Correlation Check

Correlation Check: - we used a Heatmap to view the correlation between parameters.

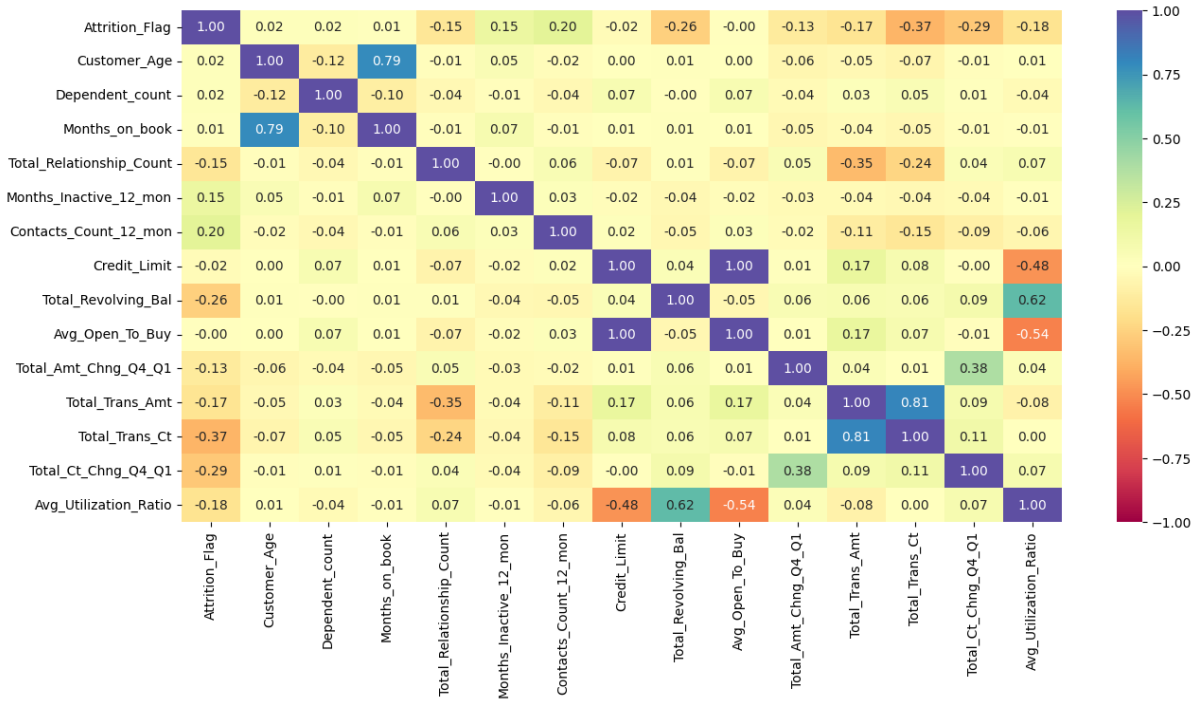

Figure 21 Correlation check

- Months_on_book and Customer_Age have a strong positive correlation (i.e 0.79)
- Avg_Open_To_Buy has a strong adverse correlation (i.e -0.54) with Avg_Utilization_Ratio.
- Total_Trans_Ct has zero (i.e 0.00) correlation with Avg_Utilization_Ratio.

5.2 Attrition_Flag vs Gender

The table shows the gender-wise attrition_flag data.

Attrition_Flag	0	1	All
Gender			
All	8500	1627	10127
F	4428	930	5358
M	4072	697	4769

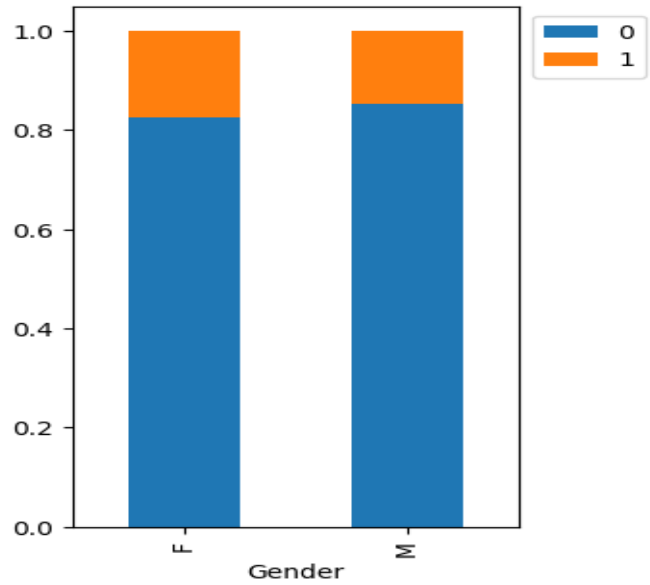

Figure 22 Attrition_Flag vs Gender

The attrition count for females is more than male gender.

5.3 Attrition_Flag vs Marital_Status

The attrition flag has been divided based on marital status.

Attrition_Flag	0	1	All
Marital_Status			
All	7880	1498	9378
Married	3978	709	4687
Single	3275	668	3943
Divorced	627	121	748

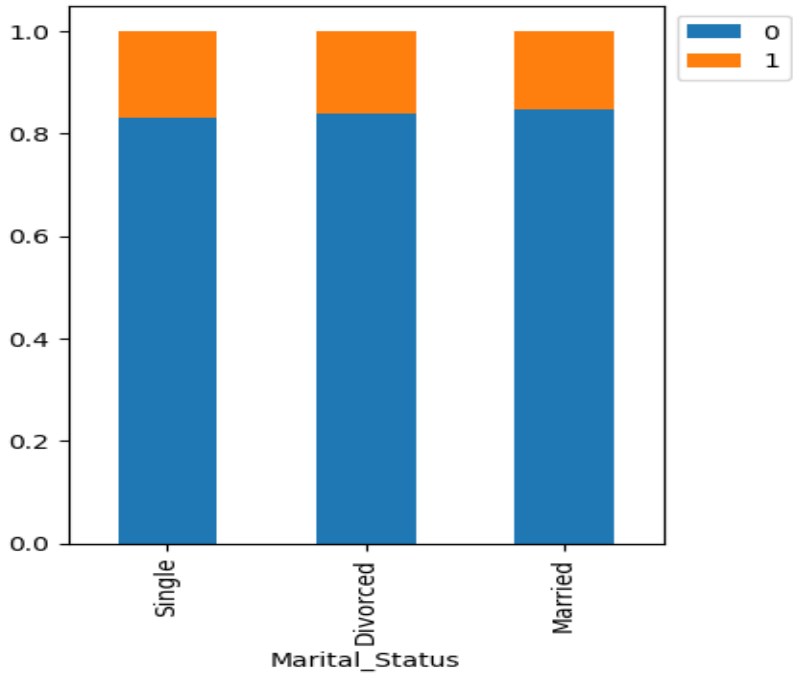

Figure 23 Attrition_Flag vs Marital_Status

5.4 Attrition_Flag vs Education_Level

The below table shows the count of status wise for Education level.

Attrition_Flag	0	1	All
Education_Level			
All	7237	1371	8608
Graduate	2641	487	3128
High School	1707	306	2013
Uneducated	1250	237	1487
College	859	154	1013
Doctorate	356	95	451
Post-Graduate	424	92	516

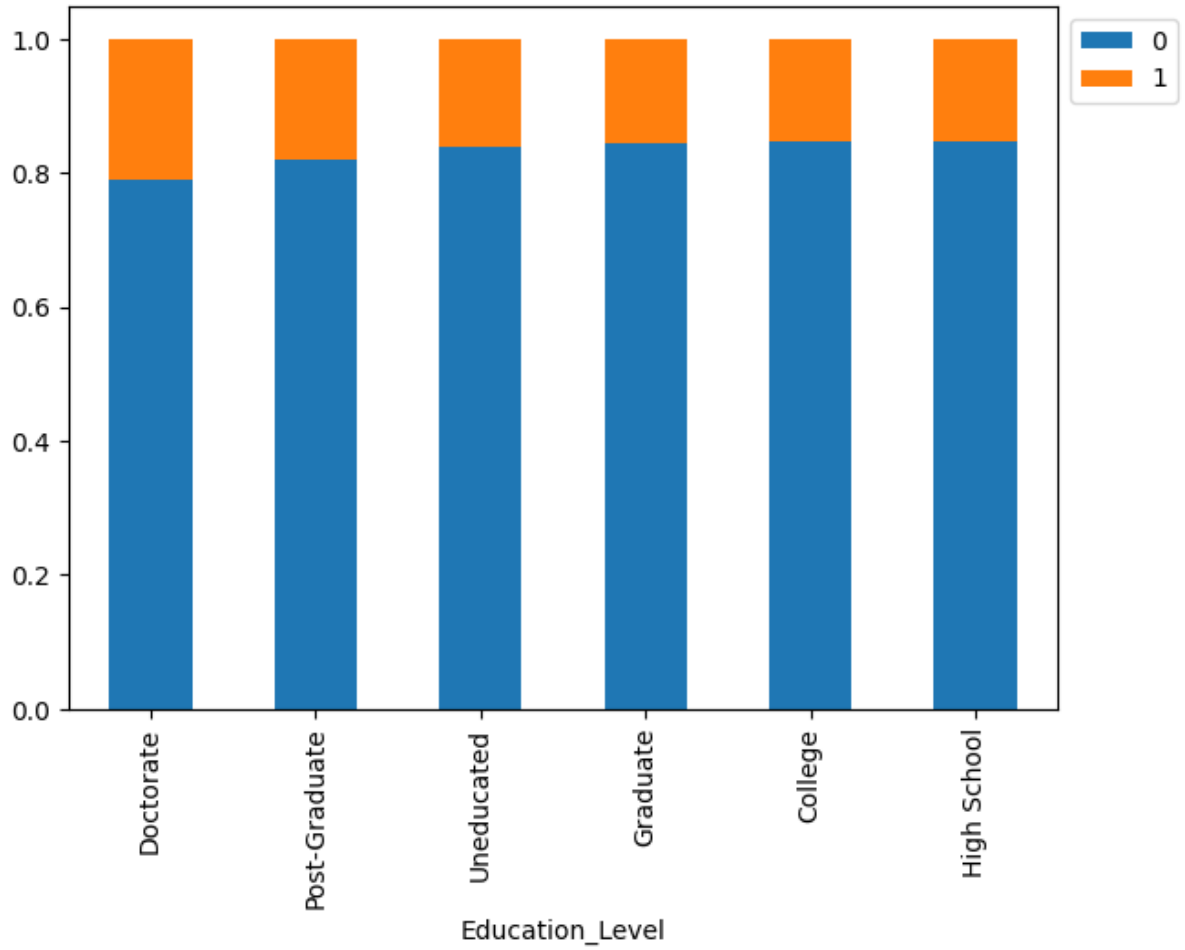

Figure 24 Attrition_Flag vs Education_Level

5.5 Attrition_Flag vs Income_Category

The below table shows the attrition flag-wise the income category.

Attrition_Flag	0	1	All
Income_Category			
All	8500	1627	10127
Less than $40K	2949	612	3561
$40K - $60K	1519	271	1790
$80K - $120K	1293	242	1535
$60K - $80K	1213	189	1402
abc	925	187	1112
$120K +	601	126	727

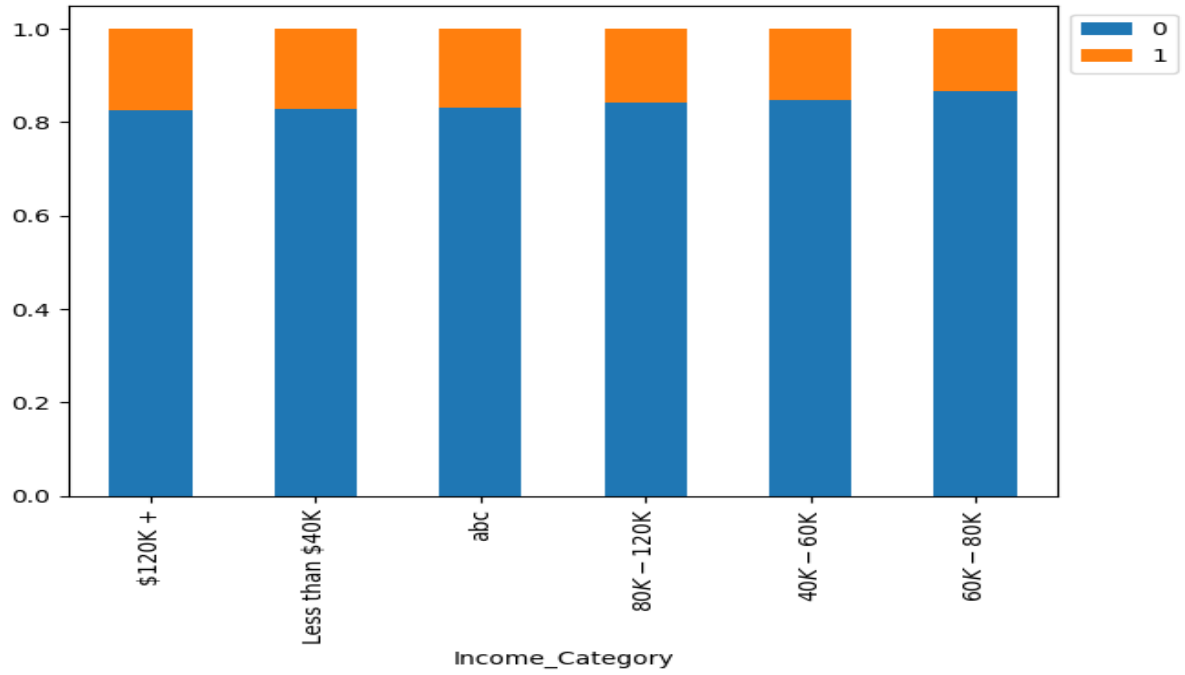

Figure 25 Attrition_Flag vs Income_Category

5.6 Attrition_Flag vs Contacts_Count_12_mon

The below table shows the attrition flag-wise for Contacts_count_12_mon.

Attrition_Flag	0	1	All
Contacts_Count_12_mon			
All	8500	1627	10127
3	2699	681	3380
2	2824	403	3227
4	1077	315	1392
1	1391	108	1499
5	117	59	176
6	0	54	54
0	392	7	399

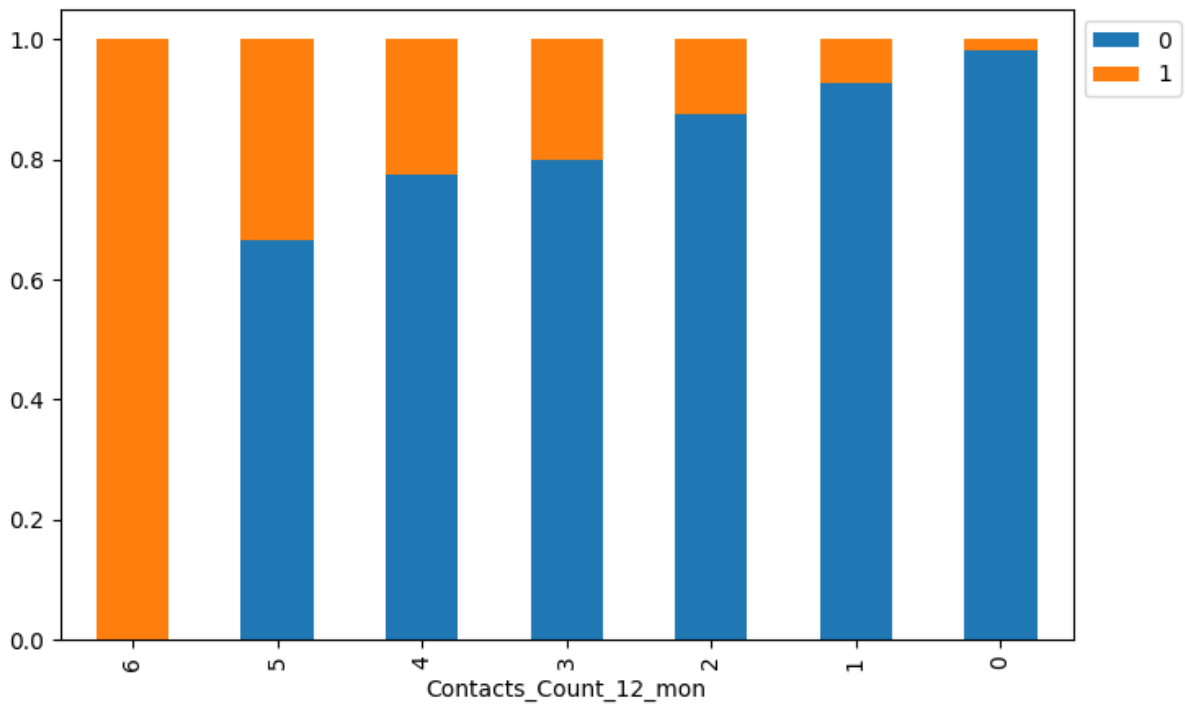

Figure 26 Attrition_Flag vs Contacts_Count_12_mon

5.7 Attrition_Flag vs Months_Inactive_12_mon

Let's see the number of months a customer was inactive in the last 12 months (Months_Inactive_12_mon) vary by the customer's account status (Attrition_Flag)

The below table shows the attrition flag Months_Inactive_12_mon.

Attrition_Flag	0	1	All
Months_Inactive_12_mon			
All	8500	1627	10127
3	3020	826	3846
2	2777	505	3282
4	305	130	435
1	2133	100	2233
5	146	32	178
6	105	19	124
0	14	15	29

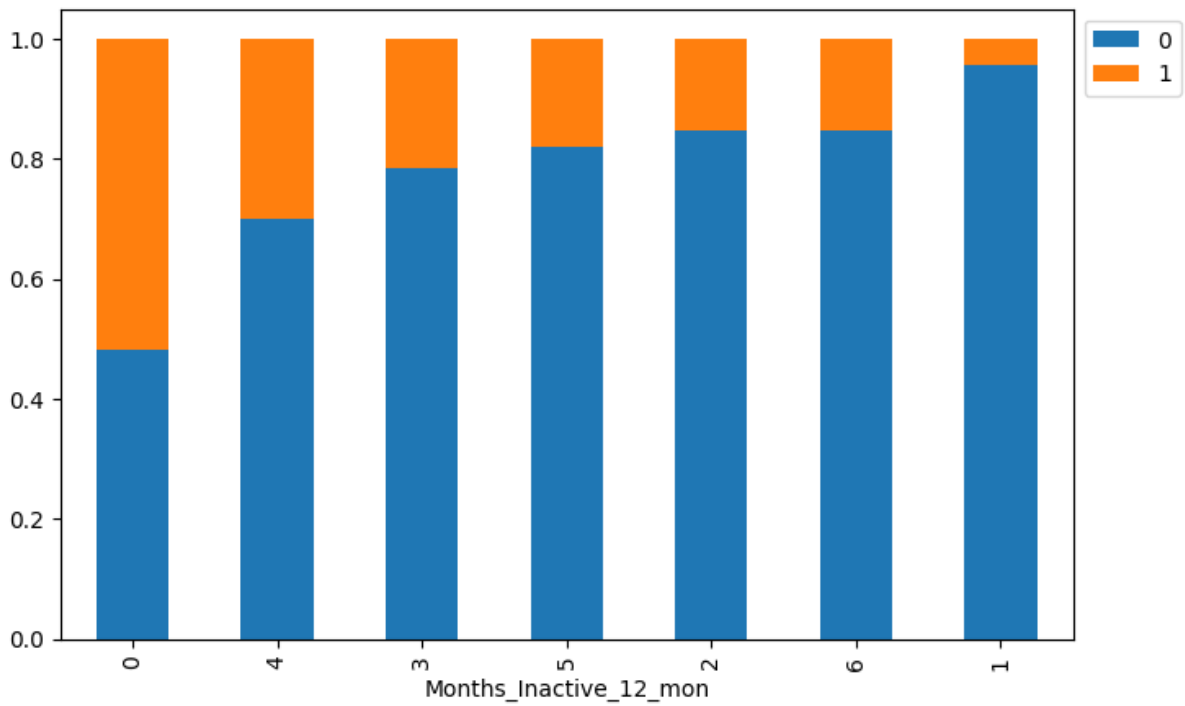

Figure 27 Attrition_Flag vs Months_Inactive_12_mon

5.8 Attrition_Flag vs Total_Relationship_Count

The below table shows the attrition flag-wise Total relationship count.

Attrition_Flag	0	1	All
Total_Relationship_Count			
All	8500	1627	10127
3	1905	400	2305
2	897	346	1243
1	677	233	910
5	1664	227	1891
4	1687	225	1912
6	1670	196	1866

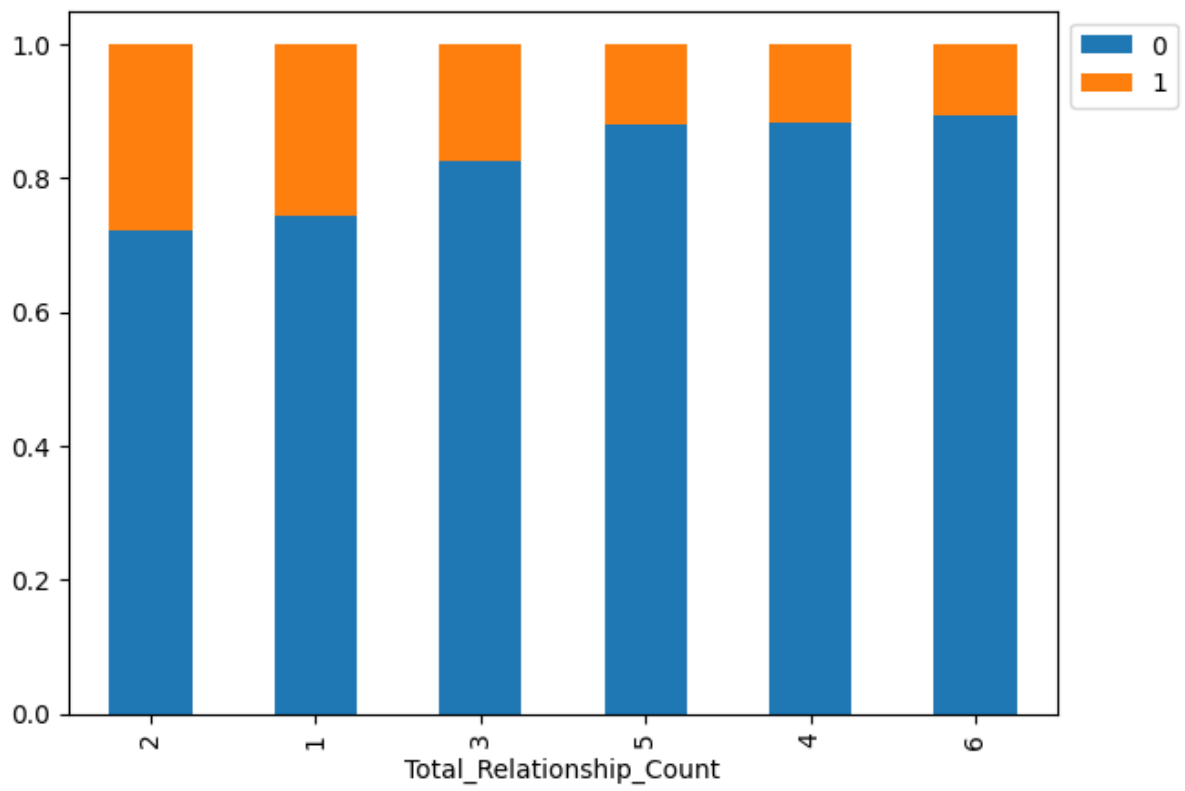

Figure 28 Attrition_Flag vs Total_Relationship_Count

5.9 Attrition_Flag vs Dependent_count

The below table shows the attrition flag wise dependent count.

Attrition_Flag	0	1	All
Dependent_count			
All	8500	1627	10127
3	2250	482	2732
2	2238	417	2655
1	1569	269	1838
4	1314	260	1574
0	769	135	904
5	360	64	424

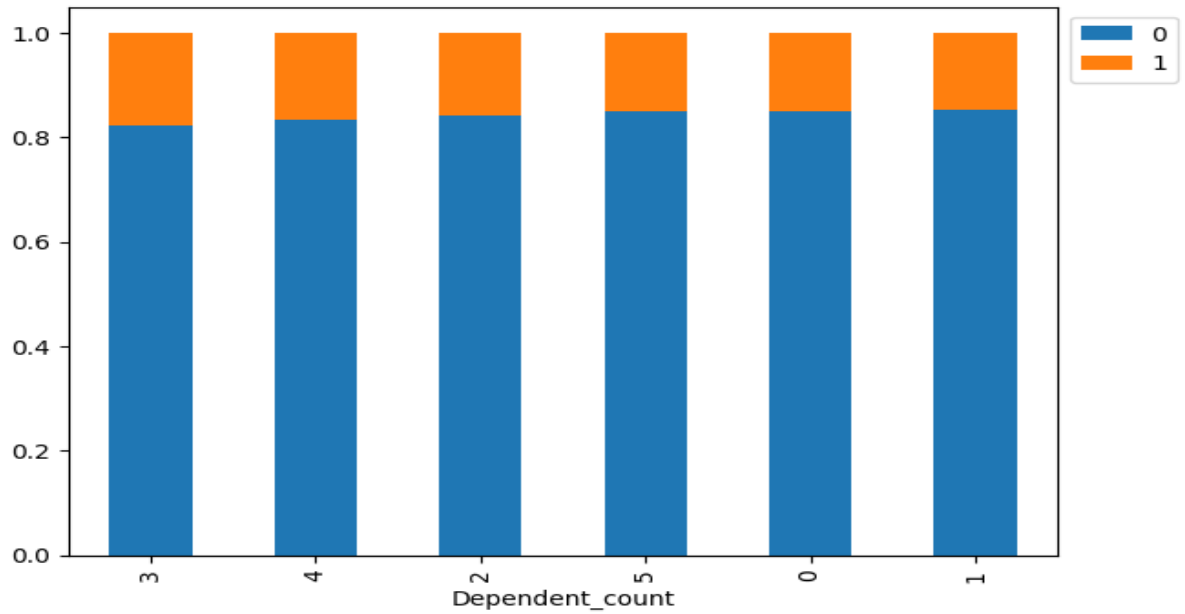

Figure 29 Attrition_Flag vs Dependent_count

5.10 Total_Revolving_Bal vs Attrition_Flag

The below table shows the attrition flag-wise total revolving bal.

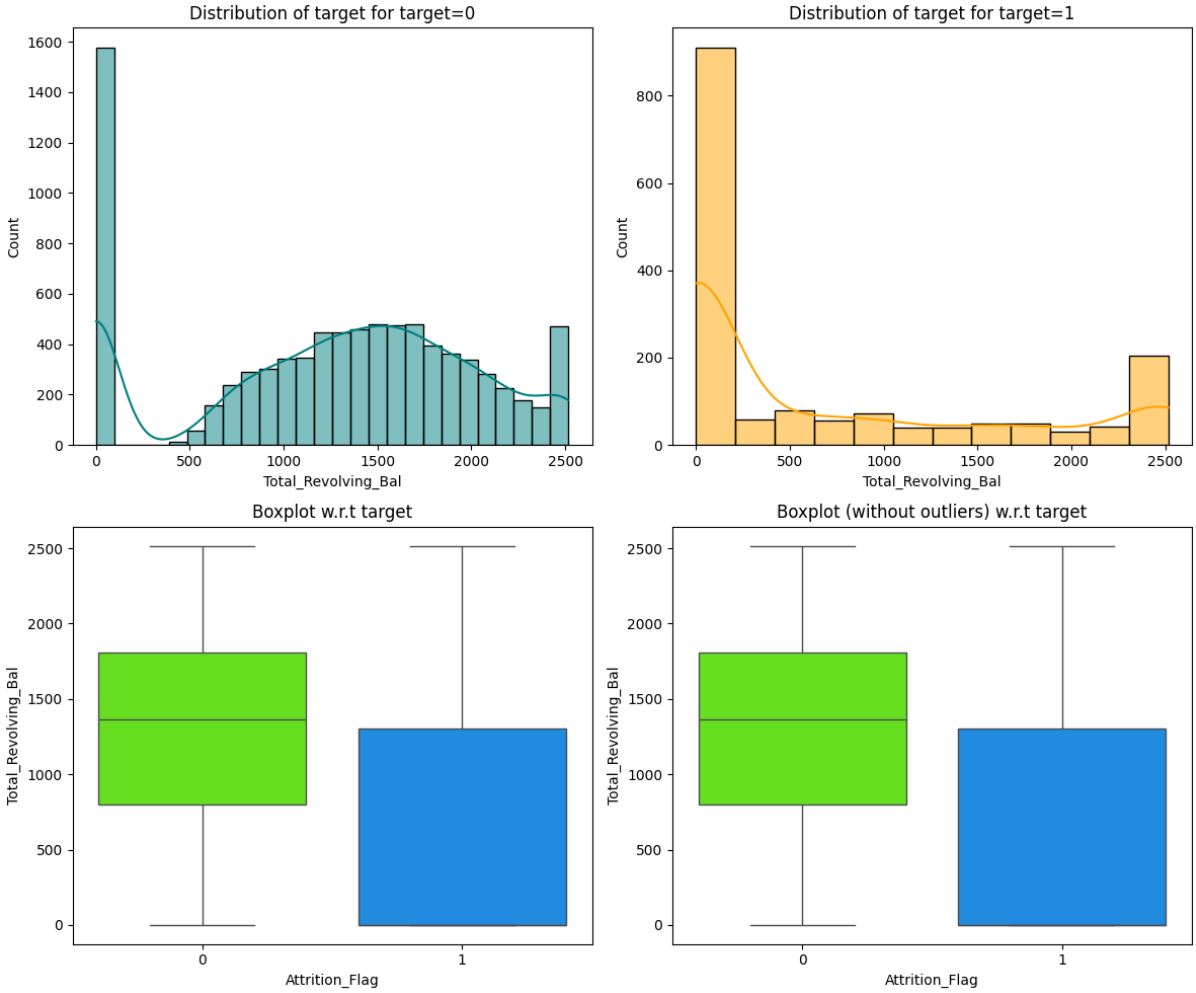

Figure 30 Total_Revolving_Bal vs Attrition_Flag

It has no outliers.

5.11 Attrition_Flag vs Credit_Limit

The below table shows the count for attrition flag wise credit limit.

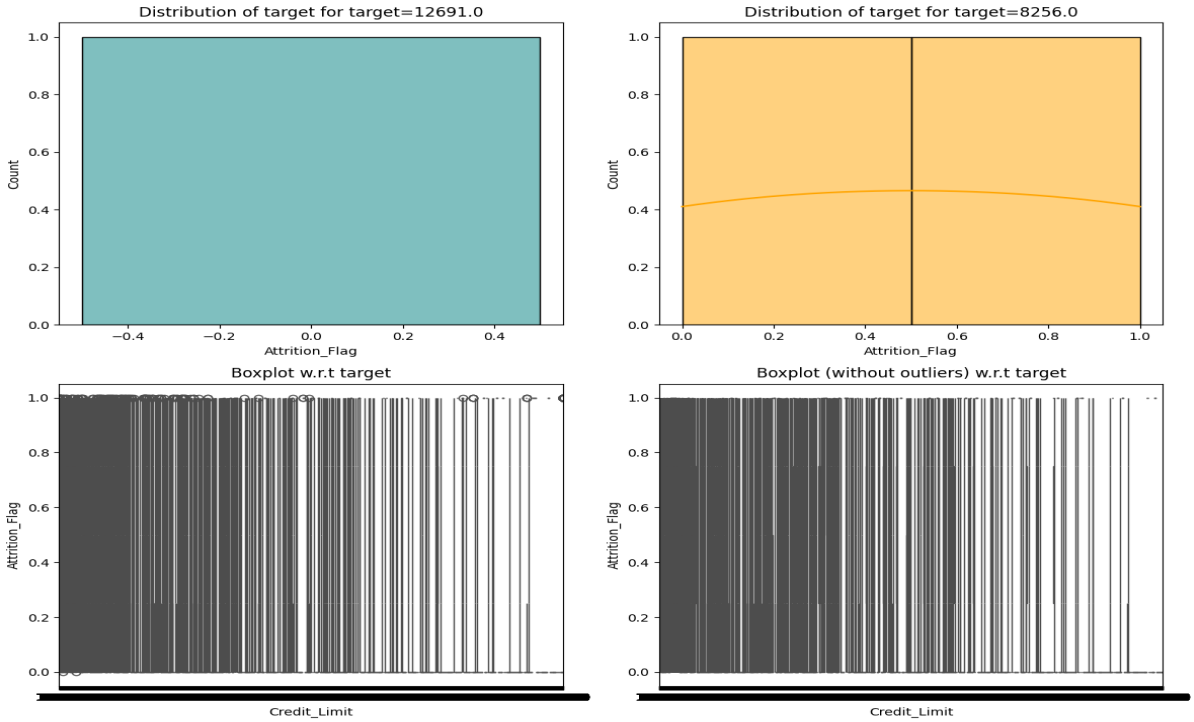

Figure 31 Attrition_Flag vs Credit_Limit

5.12 Attrition_Flag vs Customer_Age

The below table shows the attrition flag-wise customer age.

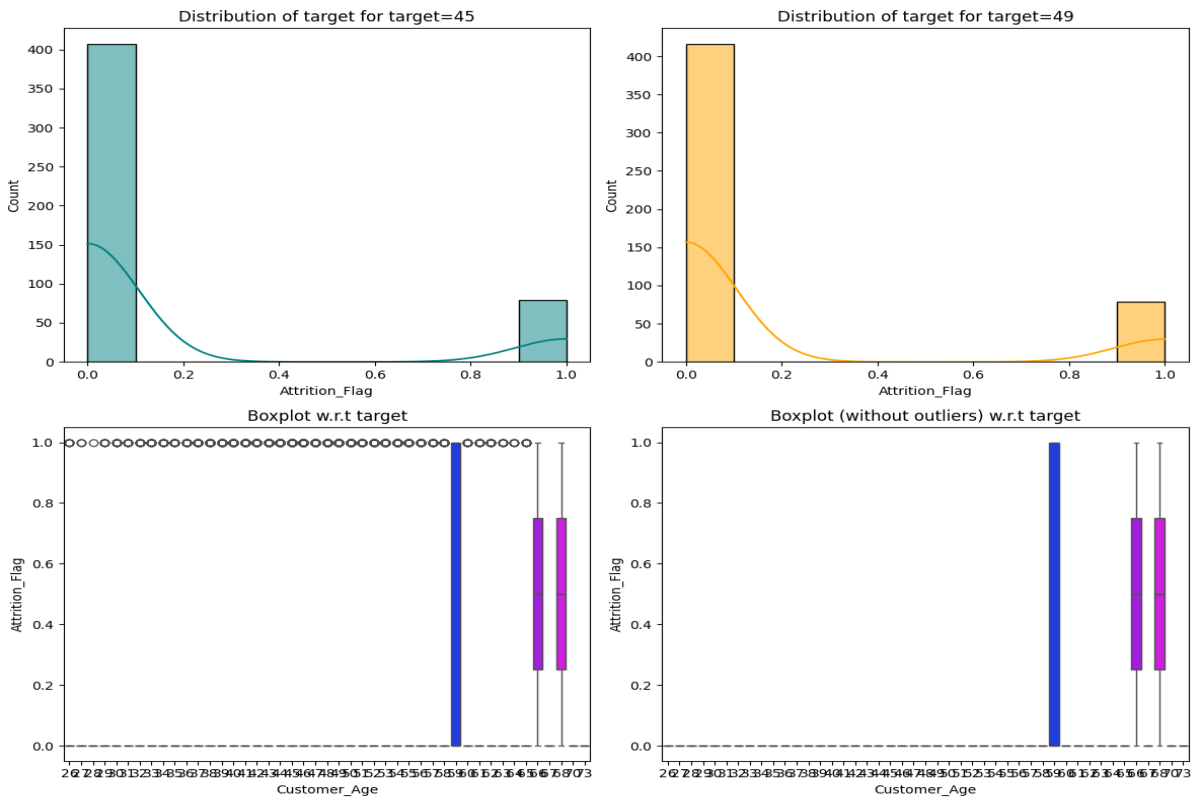

Figure 32 Attrition_Flag vs Customer_Age

5.13 Total_Trans_Ct vs Attrition_Flag

The below table shows the count of total_trans_ct and attrition flag wise total_trans_ct.

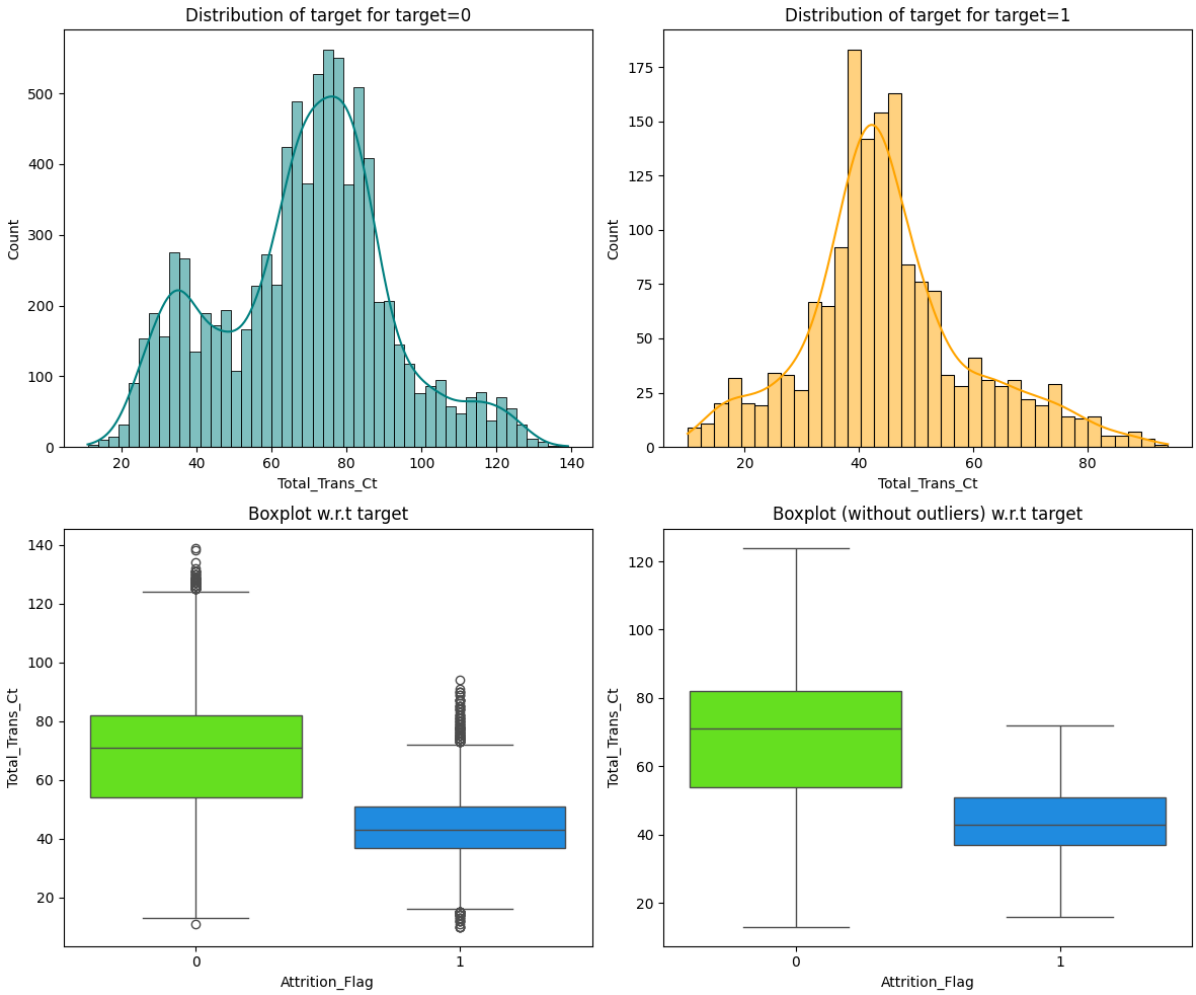

Figure 33 Total_Trans_Ct vs Attrition_Flag

It has many outliers on the upper side in boxplot w.r.t target. It has no outliers on right side of the boxplot.

5.14 Total_Trans_Amt vs Attrition_Flag

The below chart shows the attrition flag-wise total_trans_amt with outliers and without outliers.

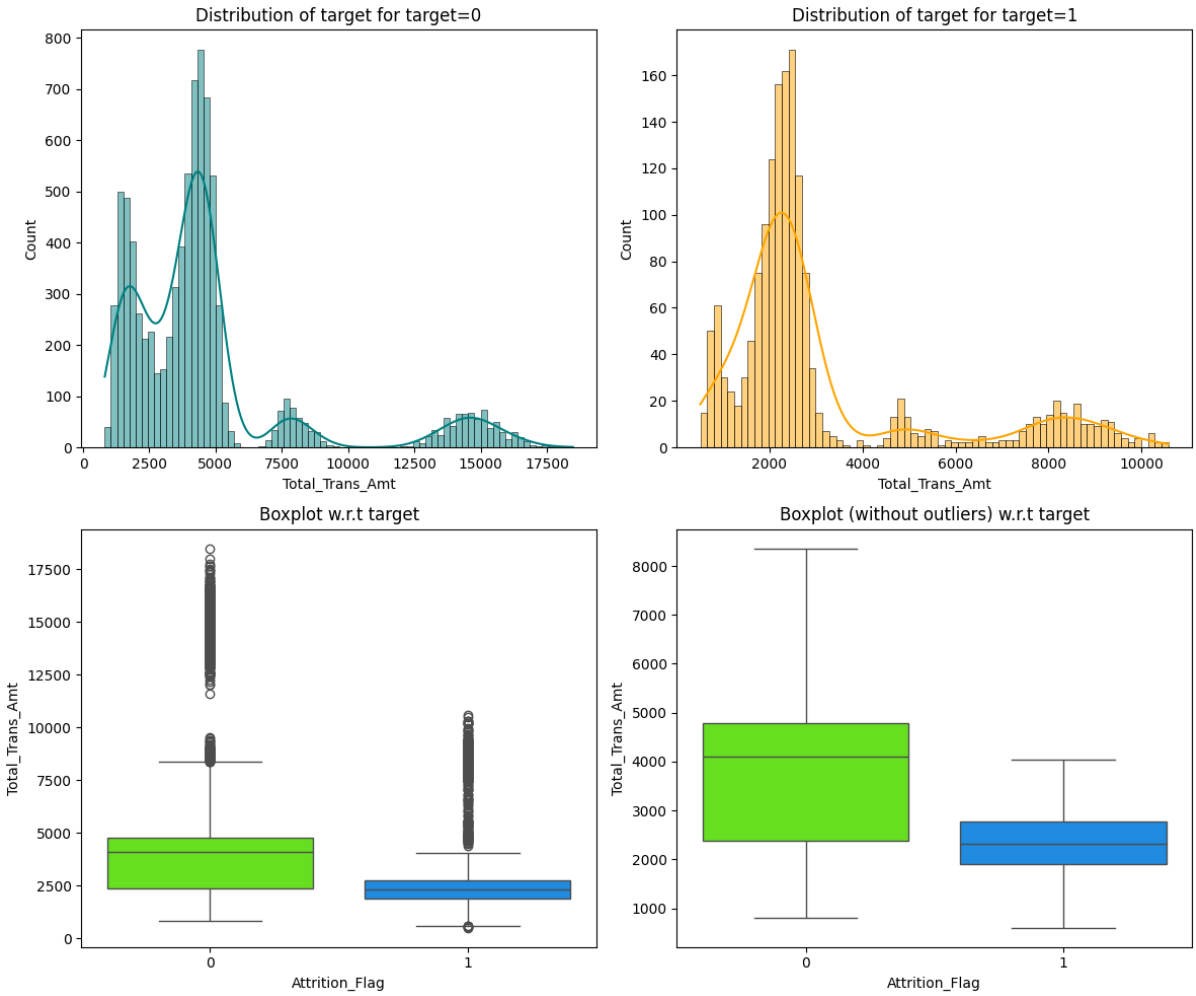

Figure 34 Total_Trans_Amt vs Attrition_Flag

5.15 Total_Ct_Chng_Q4_Q1` vs `Attrition_Flag

Let's see the change in transaction amount between Q4 and Q1 (total_ct_change_Q4_Q1) vary by the customer's account status (Attrition_Flag)

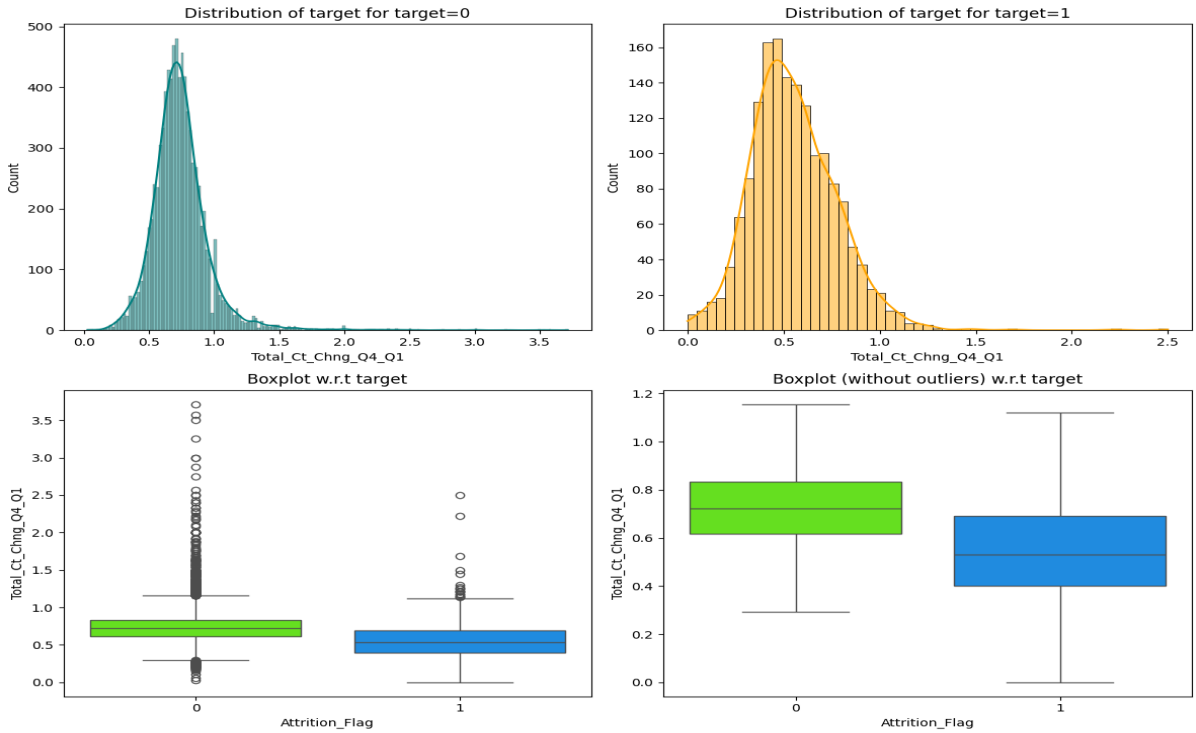

Figure 35 Total_Ct_Chng_Q4_Q1` vs `Attrition_Flag

5.16 Avg_Utilization_Ratio vs Attrition_Flag

The below chart shows the attrition flag-wise Avg_Utilization_Ratio with outliers and without outliers.

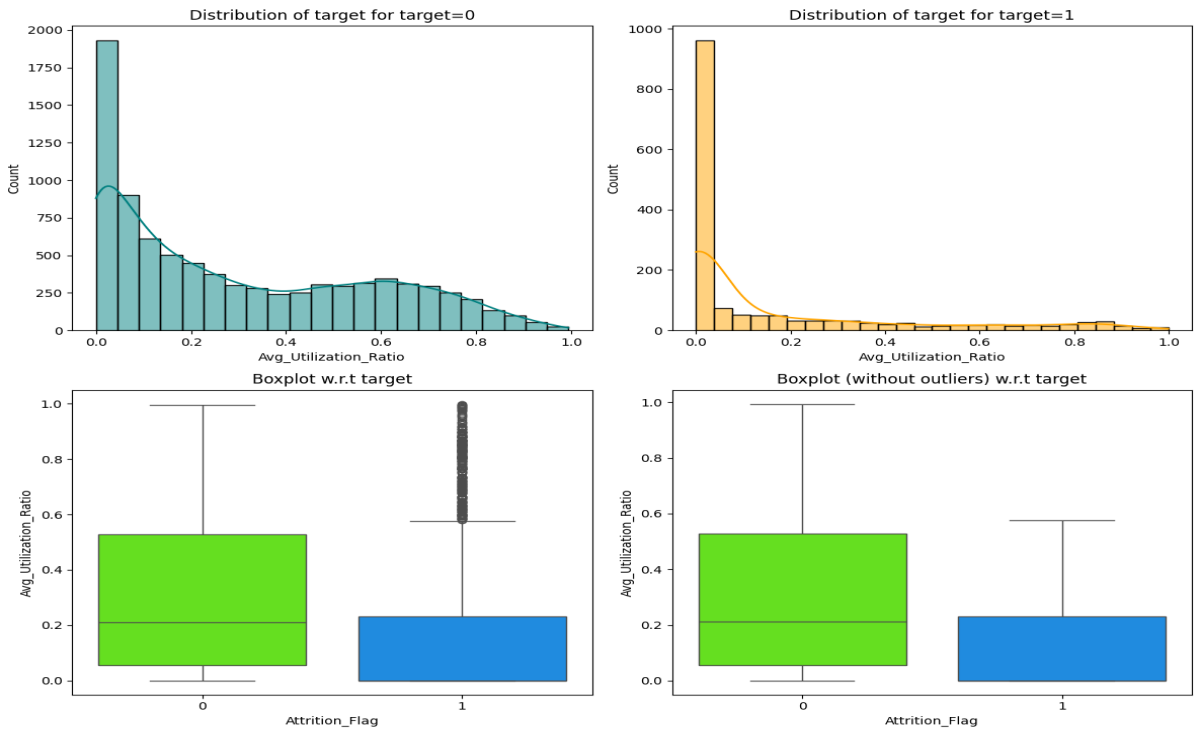

Figure 36 Avg_Utilization_Ratio vs Attrition_Flag

5.17 Attrition_Flag vs Months_on_book

The below table shows the attrition flag-wise Months on book with outliers and without outliers.

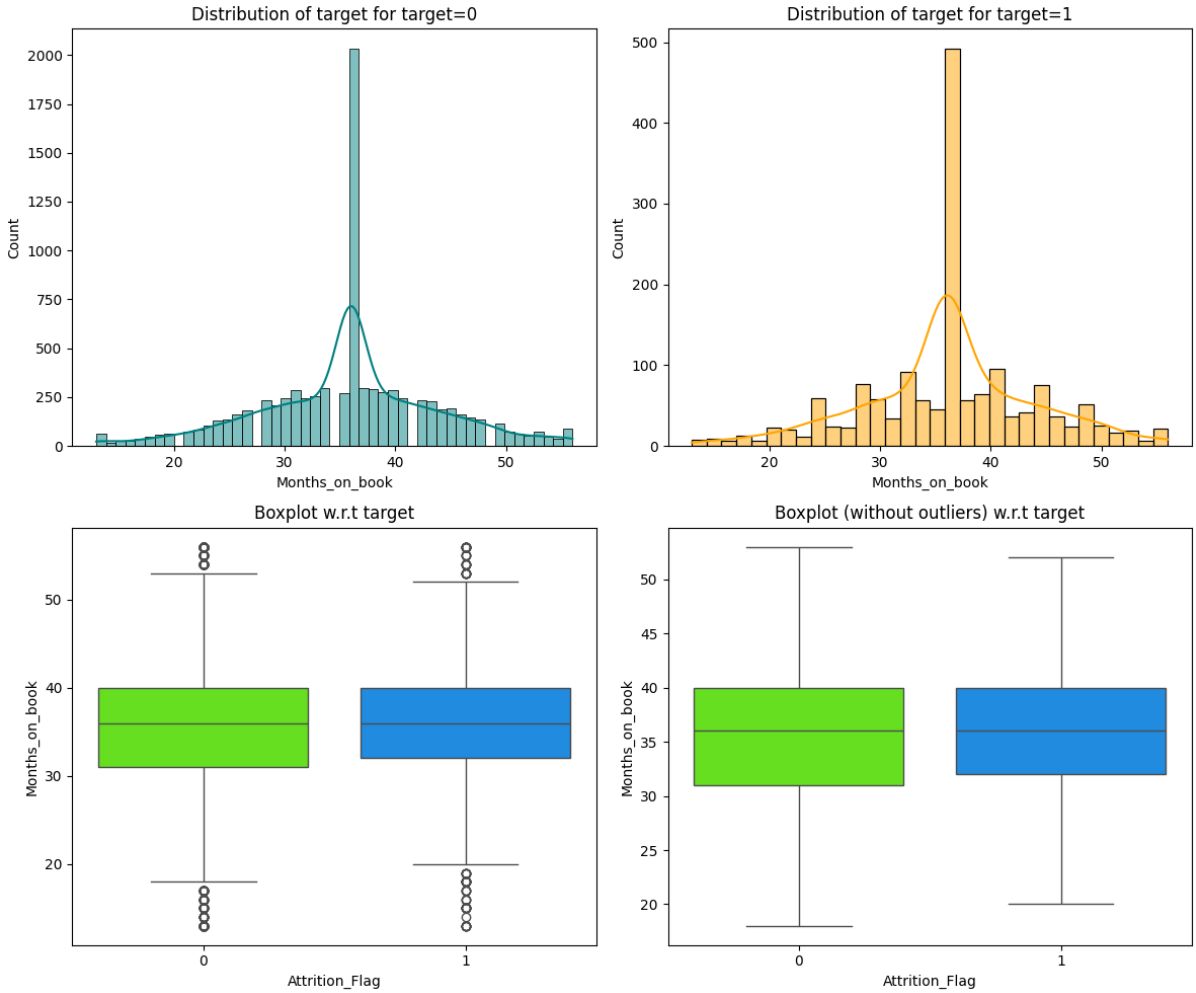

Figure 37 Attrition_Flag vs Months_on_book

5.18 Attrition_Flag vs Total_Revolving_Bal

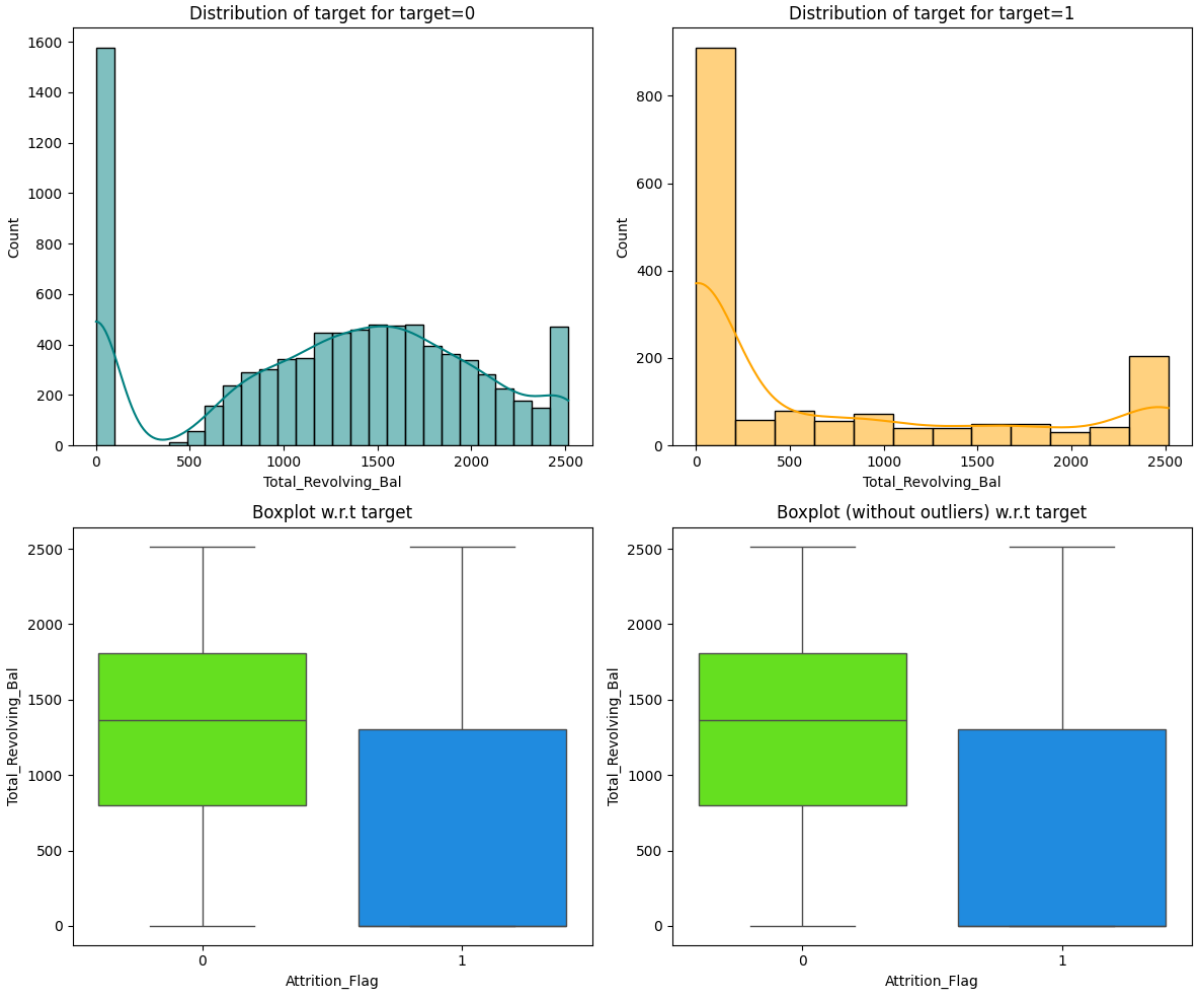

Figure 38 Attrition_Flag vs Total_Revolving_Bal

5.19 Attrition_Flag vs Avg_Open_To_Buy

The below chart shows the attrition flag-wise Avg_Open_To_Buy with outliers and without outliers.

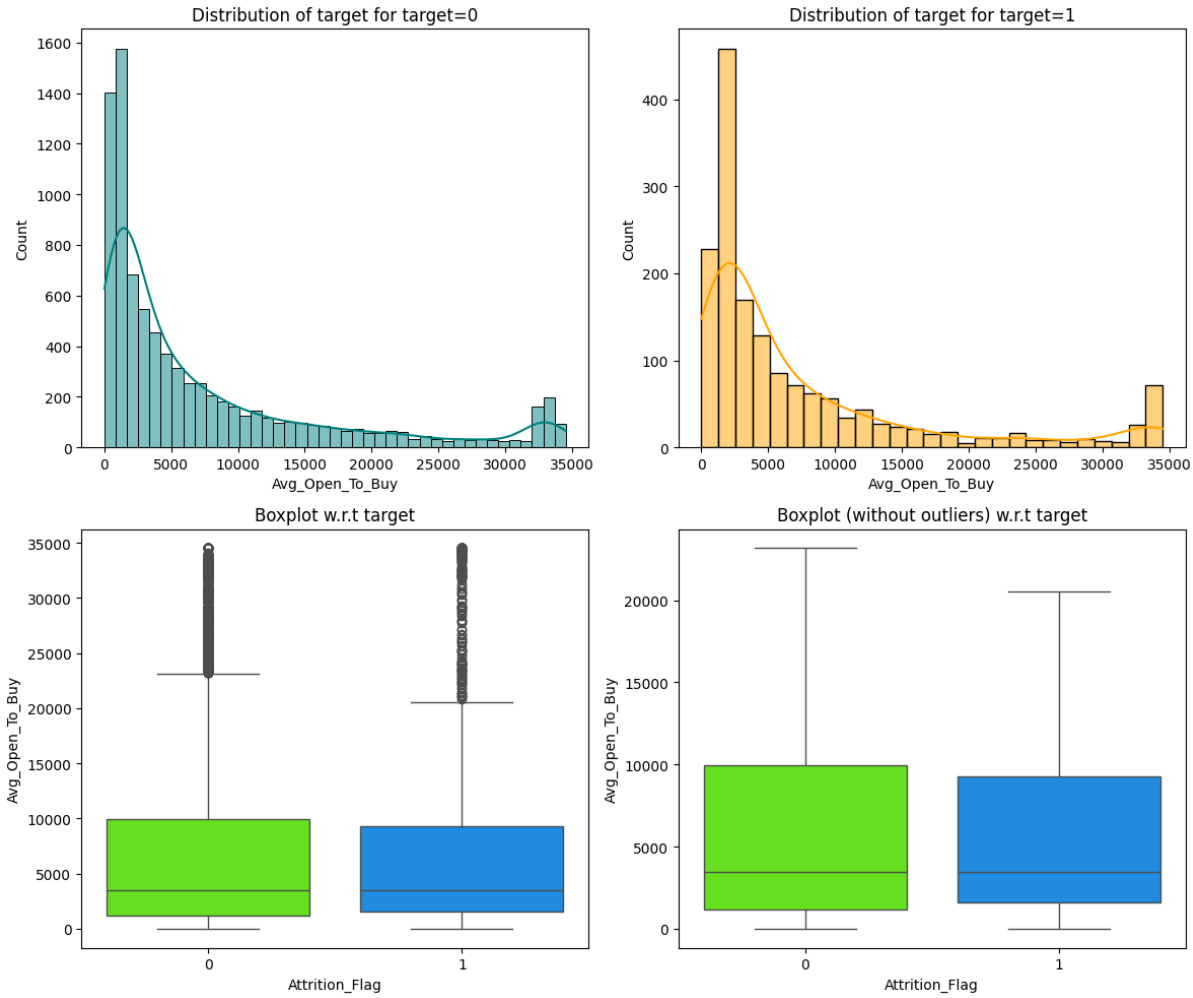

Figure 39 Attrition_Flag vs Avg_Open_To_Buy

6. Data Preprocessing

6.1 Outlier Detection:-

Checking the % outliers

	0
Attrition_Flag	16.066
Customer_Age	0.020
Dependent_count	0.000
Months_on_book	3.812
Total_Relationship_Count	0.000
Months_Inactive_12_mon	3.268
Contacts_Count_12_mon	6.211
Credit_Limit	9.717
Total_Revolving_Bal	0.000
Avg_Open_To_Buy	9.509
Total_Amt_Chng_Q4_Q1	3.910
Total_Trans_Amt	8.848
Total_Trans_Ct	0.020
Total_Ct_Chng_Q4_Q1	3.891
Avg_Utilization_Ratio	0.000

dtype: float64

Table 10 Outlier detection

6.2 Train-Test Split

The sum of unknown (i.e NaN) values for education level – 1519 & Marital status – 749.

	0
Attrition_Flag	0
Customer_Age	0
Gender	0
Dependent_count	0
Education_Level	1519
Marital_Status	749
Income_Category	0
Card_Category	0
Months_on_book	0
Total_Relationship_Count	0
Months_Inactive_12_mon	0
Contacts_Count_12_mon	0
Credit_Limit	0
Total_Revolving_Bal	0
Avg_Open_To_Buy	0
Total_Amt_Chng_Q4_Q1	0
Total_Trans_Amt	0
Total_Trans_Ct	0
Total_Ct_Chng_Q4_Q1	0
Avg_Utilization_Ratio	0

dtype: int64

Table 11 Sum of Unknown values

Splitting data into training and validation set:

(8101, 19) (507, 19) (1519, 19)

6.3 Missing value imputation

Checking that no column has missing values in train sets

```
Marital_Status              0
Income_Category             0
Card_Category               0
Months_on_book              0
Total_Relationship_Count    0
Months_Inactive_12_mon      0
Contacts_Count_12_mon       0
Credit_Limit                0
Total_Revolving_Bal         0
Avg_Open_To_Buy             0
Total_Amt_Chng_Q4_Q1        0
Total_Trans_Amt             0
Total_Trans_Ct              0
Total_Ct_Chng_Q4_Q1         0
Avg_Utilization_Ratio       0
dtype: int64
```

Table 12 Missing values imputation of train data set

Checking that no column has missing values in validation data

```
Customer_Age                0
Gender                      0
Dependent_count             0
Education_Level             0
Marital_Status              0
Income_Category             0
Card_Category               0
Months_on_book              0
Total_Relationship_Count    0
Months_Inactive_12_mon      0
Contacts_Count_12_mon       0
Credit_Limit                0
Total_Revolving_Bal         0
Avg_Open_To_Buy             0
Total_Amt_Chng_Q4_Q1        0
Total_Trans_Amt             0
Total_Trans_Ct              0
Total_Ct_Chng_Q4_Q1         0
Avg_Utilization_Ratio       0
dtype: int64
```

Table 13 Missing values imputation of validation data set

Checking that no column has missing values in test data set

```
Customer_Age                0
Gender                      0
Dependent_count             0
Education_Level             0
Marital_Status              0
Income_Category             0
Card_Category               0
Months_on_book              0
Total_Relationship_Count    0
Months_Inactive_12_mon      0
Contacts_Count_12_mon       0
Credit_Limit                0
Total_Revolving_Bal         0
Avg_Open_To_Buy             0
Total_Amt_Chng_Q4_Q1        0
Total_Trans_Amt             0
Total_Trans_Ct              0
Total_Ct_Chng_Q4_Q1         0
Avg_Utilization_Ratio       0
dtype: int64
```

Table 14 Missing values imputation of test data set

Gender count for train data

```
Gender
F    4279
M    3822
Name: count, dtype: int64
******************************
Education_Level
Graduate         3733
High School      1619
Uneducated       1171
College           816
Post-Graduate     407
Doctorate         355
Name: count, dtype: int64
******************************
Marital_Status
Married     4346
Single      3144
Divorced     611
Name: count, dtype: int64
******************************
Income_Category
Less than $40K    2812
$40K - $60K       1453
$80K - $120K      1237
$60K - $80K       1122
abc                889
$120K +            588
Name: count, dtype: int64
******************************
Card_Category
Blue        7557
Silver       436
Gold          93
Platinum      15
Name: count, dtype: int64
******************************
```

Table 15 Gender count of train data

Category wise object count of Gender for Validation data

```
Gender
F    266
M    241
Name: count, dtype: int64
******************************
Education_Level
Graduate         237
High School       94
Uneducated        84
College           49
Doctorate         24
Post-Graduate     19
Name: count, dtype: int64
******************************
Marital_Status
Married     272
Single      193
Divorced     42
Name: count, dtype: int64
******************************
Income_Category
Less than $40K    174
$40K - $60K        88
$60K - $80K        74
$80K - $120K       71
abc                62
$120K +            38
Name: count, dtype: int64
******************************
Card_Category
Blue        465
Silver       37
Gold          3
Platinum      2
Name: count, dtype: int64
******************************
```

Table 16 Gender count of validation data

Category wise object count of Gender for test data

```
Gender
F    4279
M    3822
Name: count, dtype: int64
******************************
Education_Level
Graduate         3733
High School      1619
Uneducated       1171
College           816
Post-Graduate     407
Doctorate         355
Name: count, dtype: int64
******************************
Marital_Status
Married     4346
Single      3144
Divorced     611
Name: count, dtype: int64
******************************
Income_Category
Less than $40K    2812
$40K - $60K       1453
$80K - $120K      1237
$60K - $80K       1122
abc                889
$120K +            588
Name: count, dtype: int64
******************************
Card_Category
Blue        7557
Silver       436
Gold          93
Platinum      15
Name: count, dtype: int64
******************************
```

Table 17 Gender count of test data

6.4 Encoding categorical variables

(8101, 30) (507, 30) (1519, 30)

- After encoding there are 29 columns.

6.5 check the top 5 rows from the train dataset

Row No.	Customer_Age	Dependent_count	Months_on_book	Total_Relationship_Count	Months_Inactive_12_mon	Contacts_Count_12_mon
9066	54	1	36	1	3	3
5814	58	4	48	1	4	3
792	45	4	36	6	1	3
1791	34	2	36	4	3	4
5011	49	2	39	5	3	4

Credit_Limit	Total_Revolving_Bal	Avg_Open_To_Buy	Total_Amt_Chng_Q4_Q1	Total_Trans_Amt	Total_Trans_Ct	Total_Ct_Chng_Q4_Q1
3723	1728	1995	0.595	8554	99	0.678
5396	1803	3593	0.493	2107	39	0.393
15987	1648	14339	0.732	1436	36	1.25
3625	2517	1108	1.158	2616	46	1.3
2720	1926	794	0.602	3806	61	0.794

Avg_Utilization_Ratio	Gender_M	Education_Level_Doctorate	Education_Level_Graduate	Education_Level_High School	Education_Level_Post-Graduate	Education_Level_Uneducated
0.464	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
0.334	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE
0.103	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
0.694	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
0.708	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE

Marital_Status_Married	Marital_Status_Single	Income_Category_$40K - $60K	Income_Category_$60K - $80K	Income_Category_$80K - $120K	Income_Category_Less than $40K	Income_Category_abc	Card_Category_Gold	Card_Category_Platinum	Card_Category_Silver
FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
FALSE	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	FALSE	FALSE
FALSE	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE
TRUE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE

Table 18 Top 5 rows of train data set

7. Model Building

7.1 Model evaluation criterion

Model can make wrong predictions as:

- Predicting a customer will attrite and the customer doesn't attrite
- Predicting a customer will not attrite and the customer attrites

Which case is more important?

- Predicting that customer will not attrite but he attrites i.e. losing on a valuable customer or asset.

How to reduce this loss i.e need to reduce False Negatives??

- Bank would want Recall to be maximized, greater the Recall higher the chances of minimizing false negatives. Hence, the focus should be on increasing Recall or minimizing the false negatives or in other words identifying the true positives (i.e. Class 1) so that the bank can retain their valuable customers by identifying the customers who are at risk of attrition.

Let's define a function to output different metrics (including recall) on the train and test set and a function to show confusion matrix so that we do not have to use the same code repetitively while evaluating models.

7.2 Model Building- Original Data:-

Training Performance:

Bagging: 0.9784615384615385

Random forest: 1.0

Validation Performance:

Bagging: 0.8513513513513513

Random forest: 0.7432432432432432

7.3 Model Building- Oversampled Data:-

Before Oversampling, counts of label 'Yes': 1300

Before Oversampling, counts of label 'No': 6801

After Oversampling, counts of label 'Yes': 6801

After Oversampling, counts of label 'No': 6801

After Oversampling, the shape of train_X: (13602, 30)

After Oversampling, the shape of train_y: (13602,)

7.4 Build performance models on oversampled data

Training Performance:

Bagging: 0.8918918918918919

Random forest: 0.8378378378378378

Adaboost: 0.8108108108108109

Gradient Boosting: 0.9054054054054054

Decision Tree: 0.9054054054054054

Validation Performance:

Bagging: 0.8918918918918919

Random forest: 0.8378378378378378

Adaboost: 0.8108108108108109

Gradient Boosting: 0.9054054054054054

Decision Tree: 0.9054054054054054

7.5 Model Building- Undersampled Data

Before Under Sampling, counts of label 'Yes': 1300

Before Under Sampling, counts of label 'No': 6801

After Under Sampling, counts of label 'Yes': 1300

After Under Sampling, counts of label 'No': 1300

After Under Sampling, the shape of train_X: (2600, 30)

After Under Sampling, the shape of train_y: (2600,)

7.6 build performance models on undersampled data

Training Performance:

Bagging: 0.9930769230769231

Random forest: 1.0

Adaboost: 0.9538461538461539

Gradient Boosting: 0.9784615384615385

Decision Tree: 1.0

Validation Performance:

Bagging: 0.9054054054054054

Random forest: 0.918918918918919

Adaboost: 0.9324324324324325

Gradient Boosting: 0.9594594594594594

Decision Tree: 0.918918918918919

7.7 Hyperparameter Tuning:-

7.7.1 Note

1. Sample parameter grids have been provided to do necessary hyperparameter tuning. These sample grids are expected to provide a balance between model performance improvement and execution time. One can extend/reduce the parameter grid based on execution time and system configuration.
    - Please note that if the parameter grid is extended to improve the model performance further, the execution time will increase
2. The models chosen in this notebook are based on test runs. One can update the best models as obtained upon code execution and tune them for best performance.

7.7.2 Tuning AdaBoost using original data:-

Best parameters are {'n_estimators': 100, 'learning_rate': 0.1, 'estimator': DecisionTreeClassifier(max_depth=3, random_state=1)} with CV score=0.8699999999999999:

CPU times: user 4.78 s, sys: 431 ms, total: 5.21 s

Wall time: 2min 38s

Creating new pipeline with best parameters - fit the model on original data

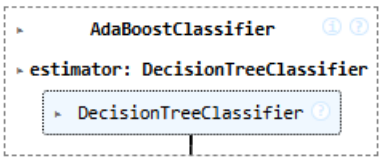

Figure 40 AdaBoost Classifier

check the performance on training set

	Accuracy	Recall	Precision	F1
0	0.985	0.934	0.969	0.951

Table 19 Adaboost performance on train data

Checking model's performance on validation set

	Accuracy	Recall	Precision	F1
0	0.966	0.892	0.880	0.886

Table 20 Adaboost performance on validation data

7.7.3 Tuning Ada Boost using undersampled data

Creating new pipeline with best parameters - fit the model on undersampled data

```
AdaBoostClassifier
estimator: DecisionTreeClassifier
DecisionTreeClassifier
```

Figure 41 Adaboost classifier – undersampled data

check the performance on training set

	Accuracy	Recall	Precision	F1
0	0.985	0.934	0.969	0.951

Table 21 Performance on train set

Checking model's performance on validation set - check the performance on validation set

	Accuracy	Recall	Precision	F1
0	0.966	0.892	0.880	0.886

Table 22 Performance on validation set

7.7.4 Tuning Gradient Boosting using undersampled data

Best parameters are {'subsample': 0.9, 'n_estimators': 100, 'max_features': 0.5, 'learning_rate': 0.1, 'init': AdaBoostClassifier(random_state=1)} with CV score=0.8392307692307692:

CPU times: user 5.16 s, sys: 568 ms, total: 5.73 s

Wall time: 3min 46s

Creating new pipeline with best parameters - the best parameters obtained from tuning

```
GradientBoostingClassifier
init: AdaBoostClassifier
AdaBoostClassifier
```

Figure 42 GradientBoosting classifier undersampled data

check the performance on undersampled train set

	Accuracy	Recall	Precision	F1
0	0.975	0.978	0.972	0.975

Table 23 Undersampled train set

check the performance on validation set

	Accuracy	Recall	Precision	F1
0	0.937	0.946	0.714	0.814

Table 24 Undersampled Validation set

7.7.5 Tuning Gradient Boosting using original data:-

Best parameters are {'subsample': 0.9, 'n_estimators': 100, 'max_features': 0.5, 'learning_rate': 0.1, 'init': AdaBoostClassifier(random_state=1)} with CV score=0.8392307692307692:

CPU times: user 4.73 s, sys: 610 ms, total: 5.34 s

Wall time: 3min 48s

Creating new pipeline with best parameters - the best parameters obtained from tuning

```
GradientBoostingClassifier
  init: AdaBoostClassifier
    AdaBoostClassifier
```

Figure 43 GradientBoosting classifier Oversmpled data set

7.7.6 Tuning Gradient Boosting using over sampled data:-

the performance on oversampled train set

	Accuracy	Recall	Precision	F1
0	0.926	0.859	0.992	0.921

Table 25 performance on oversampled train set

check the performance on validation set

	Accuracy	Recall	Precision	F1
0	0.966	0.851	0.913	0.881

Table 26 performance on validation set - oversampled

7.7.7 Tuning XGBoost Model with Original data- fit the model on original data

Best parameters are {'subsample': 0.7, 'scale_pos_weight': 5, 'n_estimators': 75, 'learning_rate': 0.05, 'gamma': 3} with CV score=0.9346153846153846:

CPU times: user 2.31 s, sys: 348 ms, total: 2.65 s

Wall time: 1min 15s

the best parameters obtained from tuning

```
XGBClassifier
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=None, device=None, early_stopping_rounds=None,
              enable_categorical=False, eval_metric='logloss',
              feature_types=None, gamma=1, grow_policy=None,
              importance_type=None, interaction_constraints=None,
              learning_rate=0.1, max_bin=None, max_cat_threshold=None,
              max_cat_to_onehot=None, max_delta_step=None, max_depth=None,
              max_leaves=None, min_child_weight=None, missing=nan,
              monotone_constraints=None, multi_strategy=None, n_estimators=100,
              n_jobs=None, num_parallel_tree=None, random_state=1, ...)
```

Figure 44 XGBClassifier

check the performance on the original train set

	Accuracy	Recall	Precision	F1
0	0.993	1.000	0.959	0.979

Table 27 XGB performance on original train set

check the performance on the validation set

	Accuracy	Recall	Precision	F1
0	0.972	0.952	0.887	0.918

Table 28 XGB performance on validation set

8. Model Comparison and Final Model Selection

8.1 Training performance comparison

Training performance comparison:

	Gradient boosting trained with Undersampled data	Gradient boosting trained with Original data	AdaBoost trained with Undersampled data
Accuracy	0.975	0.926	0.985
Recall	0.978	0.859	0.934
Precision	0.972	0.992	0.969
F1	0.975	0.921	0.951

Table 29 Training performance comparision

8.2 Validation performance comparison

```
                                        Accuracy  Recall  Precision  \
Tuned AdaBoost (Original Data)              0     0.966   0.892      0.880
Tuned AdaBoost (Undersampled Data)          0     0.966   0.892      0.880
Tuned Gradient Boosting (Undersampled Data) 0     0.937   0.946      0.714
Tuned Gradient Boosting (Original Data)     0     0.966   0.851      0.913

                                          F1
Tuned AdaBoost (Original Data)              0 0.886
Tuned AdaBoost (Undersampled Data)          0 0.886
Tuned Gradient Boosting (Undersampled Data) 0 0.814
Tuned Gradient Boosting (Original Data)     0 0.881
```

Table 30 Validation performance comparision

8.3 Check the performance on test set

```
Accuracy: 0.8973008558262014
Precision: 0.6381766381766382
Recall: 0.8853754940711462
F1 Score: 0.7417218543046358
ROC AUC Score: 0.8925297691524766
```

Table 31 test performance comparision

The data set has stable performance of tuned Adaboost for recall and precision.

Tuned Gradient boosting is not stable but is better in case of accuracy.

9. Feature Importances

check the feature importance of the best model

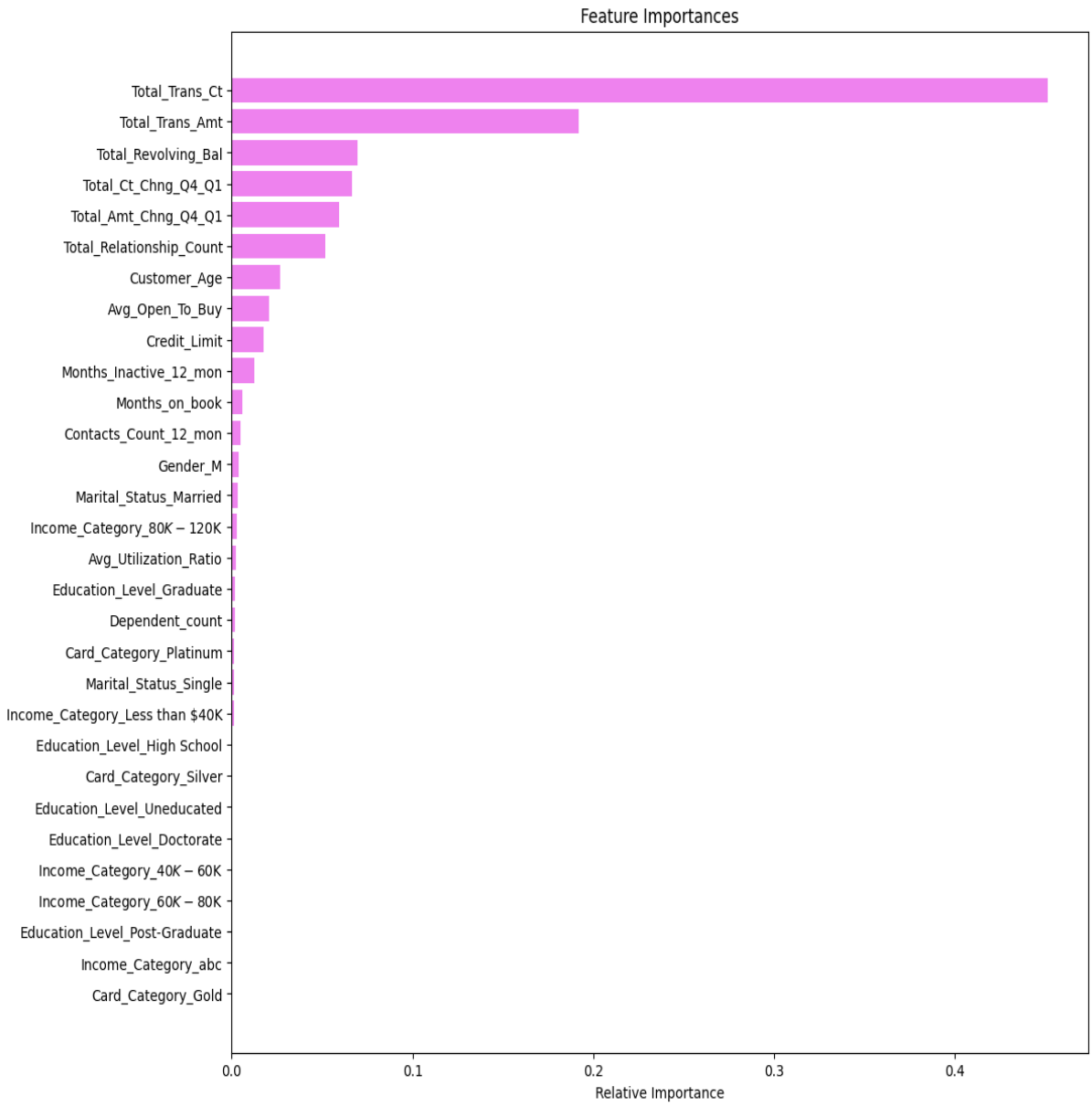

Figure 45 Feature Importances

Total_Trans_Ct has more relative importance.

Tuning the hyperparameters on Adaboost classifier has decreased in recall and better in accuracy.

Tuning the hyperparameters on Gradient boosting has increased in accuracy and decreased in recall.

10. Business Insights and Conclusions

Insights: -

- Many customers have credit limits below 5000 which increases the attrition rate.
- Customers open to buy under the 5000 so it is holding their buying power.
- Total_Amt_Chng_Q4_Q1 has too many outliers which increases the contingency in the market.
- Customers of the $120K+ income category are 727 numbers which has the possibility of expanding the market.
- Recall is not stable across all the models which leads to decrease in true positives.

Conclusions: -

- Customers must increase credit limits to satisfy their minimum needs which will help to retain the customers.
- it needs to increase the limit to only those customers to increase their buying capacity.
- Outliers have to reduce to maintain stable customers.
- Credit card benefits need to increase high-level income category people with less charges like rate of interest. It may retain the attrition customers.
- Recall needs to maintain stable to increase true positives means positive environment among the customers to reduce attrition rate.